GREENPOLE
power solutions

OUTDOOR DC POWER SYSTEM

32kW DC POWER SYSTEM

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Foreword

1. Declaration

This document provides the Green Pole Power Cabinet product introduction, safety and proper handling, Installation and commissioning and system instructions. This product should be used under the specified environment conditions specified in this document such as input voltage, operating temperature, humidity, Earthing (separate earthing for AC, DC & DG).

2. Warning Statements and Safety Symbols

DANGER	Indicates the presence of a hazard that will cause death or severe personal injury if the hazard is not avoided.
WARNING	Indicates the presence of a hazard that can cause death or severe personal injury if the hazard is not avoided.
CAUTION	Indicates the presence of a hazard that will or can cause minor personal injury or property damage if the hazard is not avoided.
	This symbol identifies the need to refer to the equipment instructions for important information
	These symbols (or equivalent) are used to identify the presence of hazardous ac mains voltage.
	This symbol is used to identify the presence of hazardous ac or dc voltages. It may also be used to warn of hazardous energy levels.
	One of these two symbols (or equivalent) may be used to identify the presence of rectifier and battery voltages. The symbol may sometimes be accompanied by some type of statement, for example: “Battery voltage present. Risk of injury due to high current. Avoid contacting conductors with un-insulated metal objects. Follow safety precautions.”
	One of these two symbols may be used to identify the presence of a hot surface. It may also be accompanied by a statement explaining the hazard. A symbol like this with a lightning bolt through the hand also means that the part is or could be at hazardous voltage levels.
	This symbol is used to identify the protective safety earth ground for the equipment.
	This symbol is used to identify other bonding points within the equipment.
	This symbol is used to identify the need for safety glasses and may sometimes be accompanied by some type of statement, for example: “Fuses can cause arcing and sparks. Risk of eye injury. Always wear safety glasses.”

The symbols may sometimes be accompanied by some type of statement, e.g., “Hazardous voltage/energy inside, Risk of injury. This unit must be accessed only by qualified personnel.”

3. General Safety Precautions

- Ensure that the product is used in environments that meet its design specifications to avoid damaging components and voiding the warranty.
- Ensure that only trained and qualified personnel install, operate, and maintain equipment.
- Comply with local laws and regulations. The safety instructions in this document are only supplements to local laws and regulations.
- Do not operate the device or cables during thunderstorms.
- Remove metal objects such as watches, bracelets, and rings when using the product.
- Use insulated tools on the product.
- Follow specified procedures during installation and maintenance.
- Measure contact point voltage with an electric meter before touching a conductor surface or terminal. Ensure that the contact point has no voltage, or it is within the specified range.
- Note that the load may power off during maintenance or fault location if the power system is not
- Connected to a battery or if battery capacity is insufficient.
- Store cables for at least 24 hours at room temperature before laying them out if they were previously stored at sub-0ºC.
- Routinely check installed equipment and perform maintenance according to the user manual; replace faulty components quickly to ensure that the device works properly.

4. Electrical Safety

Ground Connection

When installing equipment, the protective ground wire must be installed first; when removing the equipment, the protective ground wire must be removed at last. Ensure that equipment is reliably grounded before power is on.

High Voltage

	High voltage potential is present in some parts inside the power system while operational, direct, or indirect contact with these objects through the wet parts will bring fatal danger. AC power line connection may cause a short-circuit, fire or electric shock accident. AC cable connection and the area wire covered must follow the relevant regulations and design specifications. People who performed high voltage operations must have high voltage, AC, and other operations qualifications.

- Obey relevant industry safety standards and design specifications during equipment installation or operation.
- Put up "No Operation" signage on the device or switch can’t be operated during the installation process.

Tools

	Use special tools instead of common or homemade tools during high voltage and AC operation.

Short Circuit

Never make DC positive pole and DC negative pole short circuit or non-grounded pole (terminal) shorted to ground in power system during operation.

The power equipment is constant voltage DC power equipment, a short circuit will cause strong electric arcs or fire, endangering the safety of persons and equipment.

- Check the cables and polarity of interface terminals strictly during DC hot-line operations.
- Never wear watches, bracelets, rings or other conductive accessories when operating.
- Insulated tools must be used when installing and operating.
- Pay attention to the effect on the hot-line work generated by operating space, furnish the protective insulation in the operating area to avoid a short circuit dangerous caused by tool stuck or tool fall off when necessary.

Thunderstorm

Non-high voltage, AC jobs in thunderstorms.

- The atmosphere will produce a strong electromagnetic field in a thunderstorm. To avoid damage to equipment caused by lightning, you should ensure that the equipment has good grounding.

Static Electricity

Static electricity produced by the body can damage static-sensitive components on the circuit board, such as IC. Before touching the device, circuit board or IC chips, wear an ESD wrist strap and be sure the other end of ESD wrist strap is well grounded to prevent electrostatic damage to sensitive body components.

- When holding the veneer, you must hold the parts without components on the edge of the veneer and do not touch the chip with your hands.
- Removed veneer must be packed with anti - static packaging after storage or transport.

Liquid Proof, Explosion Proof

This product should be placed away from the liquid area, do not install in the vent, air conditioning, machine room outlet window and other prone to water leakage position below, prevent liquid into the equipment caused by short circuit. It is strictly prohibited to put the equipment in flammable, explosive, corrosive gas or smoke environment, and any operation in such environment is strictly prohibited.

- Do not use water to flush internal and external electrical components of equipment.
- If water, moisture, or condensation is found in the cabinet, please turn off the power immediately. When operating in a humid environment, should strictly prevent water and other conductive liquid into the equipment.

Battery Safety

	Before installing, operating, and maintaining batteries, read the instructions provided by the battery vendor. For more safety precautions, see the instructions provided by the battery vendor.

- The battery environment should be free from direct sunlight or rain, dry and well ventilated, without corrosive gas, and away from fire and organic solutions.
- Excessive battery temperature can lead to battery deformation, damage or electrolyte overflow.
- Battery has high electrical energy; irregular operation will result in danger. Follow the safety precautions required by battery operation, pay attention to the effect on hot-line work generated by operation space, take precautions to prevent battery short circuit strictly during operation.
- Battery in the handling process should always keep the electrode front up, do not invert, tilt.
- Take precautions, use protective glasses, rubber gloves, rubber boots and rubber aprons and so on.
- Incorrect handling of batteries causes hazards. When operating batteries, avoid battery short circuits and electrolyte overflow or leakage.
- Electrolyte overflow may damage the device. It will corrode metal parts and circuit boards and ultimately damage the device and cause short circuits of circuit boards.
- When installing and maintaining batteries, pay attention to the following points:
- Use special insulation tools. Take care to protect your eyes when operating batteries.
- Wear rubber gloves and a protective coat in case of electrolyte overflow.
- When handling a battery, ensure that its electrodes are upward. Leaning or reversing the battery is prohibited.
- Switch off the power supply during installation and maintenance.
- Secure battery cables to a torque specified in battery documentation. Loose connections will result in excessive voltage drop or cause batteries to burn out when the current is large.
- Short circuits will generate high transient currents and release a great deal of energy, which may cause personal injury. If conditions permit, disconnect the batteries in use before performing any other operations.
- Do not use unsealed lead-acid batteries.
- Place and secure lead-acid batteries horizontally to prevent device inflammation or corrosion due to flammable gas emitted from batteries. Lead-acid batteries in use emit flammable gas. Therefore, store the batteries in a place with good ventilation, and take measures against fire.
- High temperatures may result in battery distortion, damage, and electrolyte overflow. When the battery temperature is higher than 60°C, check the battery for electrolyte overflow. If the electrolyte overflows, absorb and counteract the electrolyte immediately. When moving or handling a battery whose electrolyte leaks, exercise caution because the leaking electrolyte may hurt human bodies.
- When you find electrolyte leaks, use sodium bicarbonate ($NaHCO_3$) or sodium carbonate (Na_2CO_3) to counteract and absorb the leaking electrolyte.

- After you connect batteries, ensure that the battery fuse is disconnected, or the circuit breaker is OFF before powering on the power system. This prevents battery over discharge, which damages batteries.

Mechanical Safety

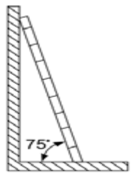
Hoisting Devices

- Only trained and qualified personnel can perform hoisting operations.
- Before hoisting objects, ensure that hoisting tools are firmly fixed onto a weight-bearing object or wall.
- Ensure that the angle formed by two cables is less than 90 degrees.

Using a Ladder

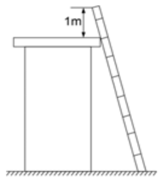
- Use only ladders that are in good condition. Find out and do not exceed the maximum weight capacity.
- The recommended angle for a ladder against another object is 75 degrees. Measure the gradient with a right angle or your arms, as shown in Figure 1-1. Ensure that the wider end of the frame is at the bottom, the base cannot slide, and that the ladder is securely positioned.

Figure 1-1 correct angle for ladders



When climbing a ladder:

- Ensure that your body's center of gravity does not shift outside the legs of the ladder.
- To minimize the risk of falling, steady your balance on the ladder before performing any operation.
- Do not climb higher than the fourth rung from the top.
- To climb onto a roof, ensure that the ladder top is at least one meter higher than the roofline, as shown in Figure 1-2.



Moving Heavy Objects

- Be cautious to prevent injury when moving heavy objects.
- Wear protective gloves when moving heavy objects.

Other Considerations:

Acute angle of objects

	It is strictly prohibited to modify the structure of equipment and change the position of plate parts without authorization. It is strictly forbidden to change components without authorization and drill holes on the cabinet without authorization. Non-compliant changes will affect the heat dissipation of equipment, electromagnetic shielding and other performance, self-drilling will also lead to metal chips into the cabinet resulting in short circuit board.

Object Angle

	Wear protective gloves when handling equipment or removing the equipment packing case with hands to prevent it from being cut.

5. Delivery & Technical Support Helpline

When delivered, inspect the package integrity and the physical condition of the cabinets carefully.

In case of any damage sustained during transport, immediately inform the carrier, and contact your local Service Center. A detailed report of the damage is necessary for any insurance claim.

⚠	NOTE! A DAMAGED CABINET MUST NEVER BE INSTALLED AND COMMISSIONED

In case of any support in regard of Installation & Commissioning and operation troubleshooting contact following no. and email.

Technical support helpline no's:	Level-1: +223 93 41 17 85
	Level-2: +223 76 51 62 34
	Level-3: +91 9445521431
	e-mail: support@greenpole-ps.com

6. Instructions & Installation Requirements

6.1. Dos and Don'ts

Dos	Don'ts
Read and understand this "User Manual"	Do not dismantle the Power Cabinet and battery bank without authorized service Engineer
Always use Insulated tools only for any field activity and Follow PPE during all site related activity.	Do not place metallic objects on Power Cabinet shelf and battery.
Keep the battery bank away from the metallic objects, Sparks, Flames and smoking materials.	Do not use any physically damaged Power Cabinet Unit and Battery
Follow Installation and commissioning steps as specified in the manual. Make sure of grounding connection first connected while installation and last removed while decommissioning.	Don't Leave the ground wire open
Power Cabinet Unit and Batteries to be service in authorized centre only	Do not short circuit the AC and DC power terminals.
Check cabinet cooling fans working & ensure enough gap Infront and rear of the system for air ventilation.	Do not use battery beyond 80% DOD or discharged battery. Charge them externally and then install.
Use isolated RS485 to USB converter only. Ensure the Slave/ Master to RS485 port are isolated before connecting with GP System RS485 Port	Do not throw water on Power Cabinet Unit and battery terminals.
Use shielded cable for connecting RS485 ports	Charge the battery in a cool area having ambient temperature less than 45 °C.
After I&C completed close & lock the system doors properly & handover the key to customer representative with sign off document duly signed.	Don not Leaves equipment exposed to rain or dusty environment by keeping its door open.
Use LVD force on switches in case of emergency only & Operate LVD Force on switches as per instruction	Do not keep LVD force on switches in ON position always

Use MCT Cable manager for cable entry & close unused cable entries & seal with silicone sealant Store the removed CS layers safely and can be used in case if required in future	Do not keep open the cable entry holes & do not through the removed CS module layers.

6.2. Installation & commissioning Requirements

Only trained and qualified construction personnel are allowed to perform the installation.

CAUTION

Our company will not be liable for the damage to devices or injury to humans caused by failure to follow instructions in this document.

Before installation, pay attention to the following items:

The customer's technical engineers must be trained and be familiar with the proper installation and operational methods. The number of installation personnel varies based on the project progress and the installation environment. Typically, two to four people are required.

Tools Requirements

CAUTION **NOTICE**

Use tools with insulated handles!

Introduction of tools and instrumentation (including, but not limited to, use of the following table tools) required before installation operations are performed.

Table 6.1 Tools and meters

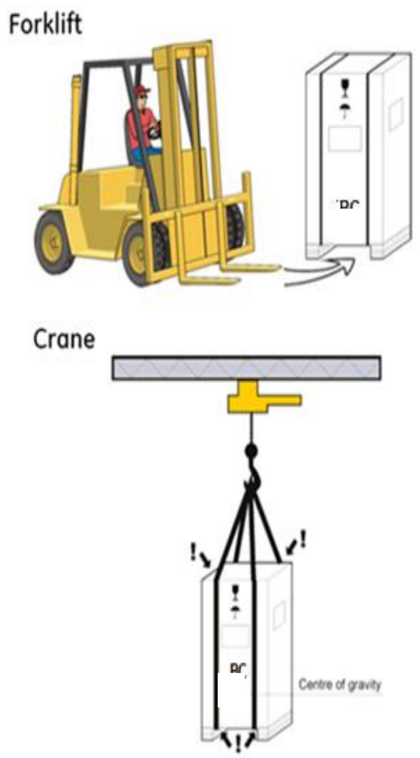
Tools and Meters				
Pallet truck 	Ladder 	Hammer drill and drill bit 	Utility knife 	Heat gun 
Rubber mallet 	Insulated adjustable wrench 	Multimeter 	Flat-head screwdriver 	Phillips screwdriver 
Ruler 	Insulated torque wrench M5 to M12 	Cable tie 	Wire nippers 	Insulation gloves 
Polyvinyl chloride (PVC) insulation tape 	Combination wrench 	Heat shrinks tubing 	Crimping pliers 	Insulation protective shoes 
Hydraulic pliers 	Diagonal pliers 	Wire stripper 	Measuring tape 	Marker 

6.3. Requirements for Cable Routing

- Cables should be more than 50mm away from heat sources to prevent insulation layer damage (melting) functional degradation (aging or breakage).
- The bending radius of cables should be at least five times the diameter of the cables.
- Cables of the same type should be bound together. Cables of different types should be a minimum of 30 mm from each other to avoid tangling.
- Cables that are bound together should be close to each other, tidy, and undamaged.
- Ground cables must not be bound to or tangled with signal cables. There should be an appropriate distance between them to minimize interruptions.
- AC power cables, DC power cables, and communications cables must be bound separately.
- Power cables must be routed straight. No joints or welding should be performed on power cables.
- Use a longer cable if necessary.

7. Transportation, Unloading and Unpacking

- The Outdoor Power Cabinet is packed on a suitable pallet to be handled by a forklift or a crane
- Pay attention to the center of gravity
- Do not tilt the cabinets more than +/- 5 degrees
- Move the Outdoor power Cabinet in its original package until its destination for installation
- Do not stack other packages on top of the cabinet. It could damage it
- The Cabinet should be strapped and buckled during transportation show that it won’t topple or fall from the carrier.
- If the battery cabinet is to be lifted by a crane, use suitable lifting straps and separator bars
- Take all the necessary precautions to avoid damage to the cabinet for loading and unloading from the carrier.
- Once unloaded position the cabinet on the uniform plinth. Now start unpacking the carton and inspect the cabinet condition if any damage observed inform the authority to take necessary action. In any case damaged cabinet can be installed at the site.



8. Product Overview

GreenPole Outdoor Power Cabinet adopts a modularized design and combined structure. It consists of parts such as the controller, rectifier module, AC distribution unit and DC distribution unit.

GP-24KW Outdoor Power system is mainly used in areas with no or poor Grid power supply, to provide power supply for small-sized station equipment, through the intelligent management of energy and battery, adjust the working mode to provide the most economical solution.

GP-24KW Outdoor Power Cabinet converts the AC to DC into stable direct current, which is applicable to the Telecom applications range.

9. Overall Product Dimension and Grouting Detail

System dimensions from all sides and grouting (mounting) details shown in Figure 9.1

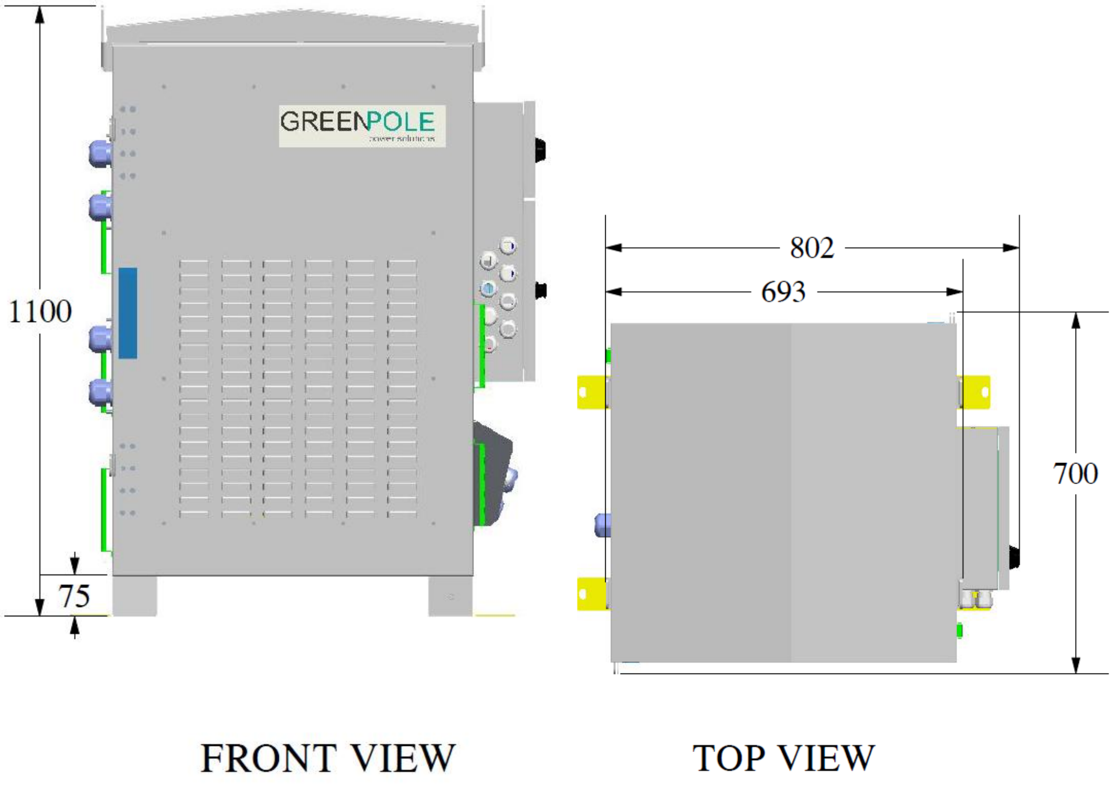


Figure 9.1 Dimension of Full system Figure

9.1. System Grouting Installation

a. Prepare the Plan surface area with concrete bed for installation
b. Mark on the concrete bed as per system grouting dimensions (Fig 9.1.1)

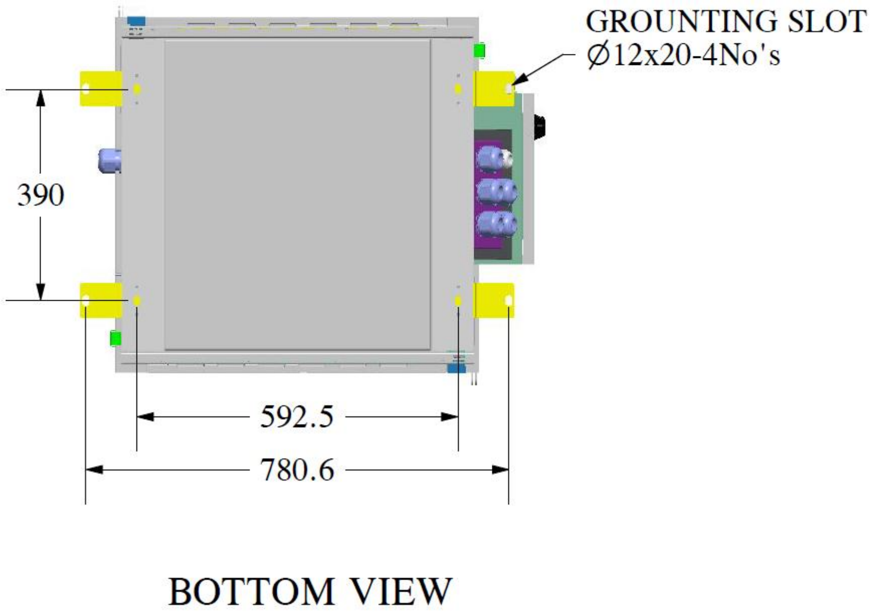
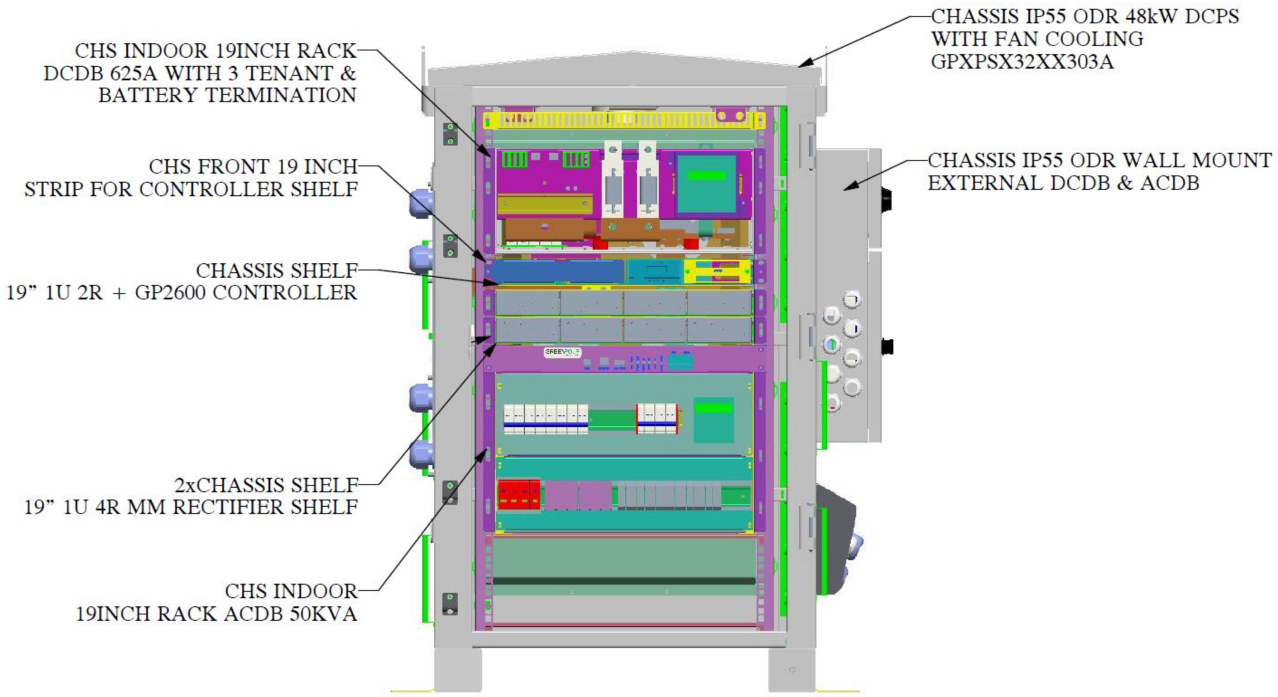


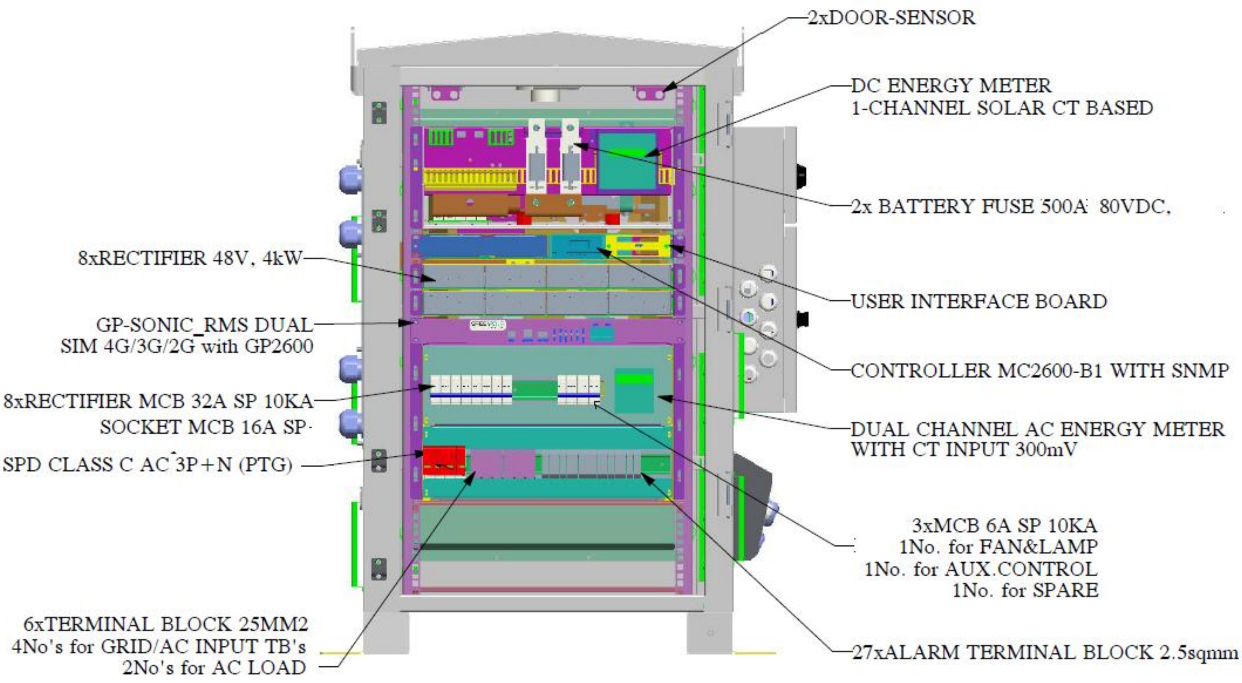
Figure 9.1.1 system Grouting dimensions

9.2. Mechanical Assembly Details

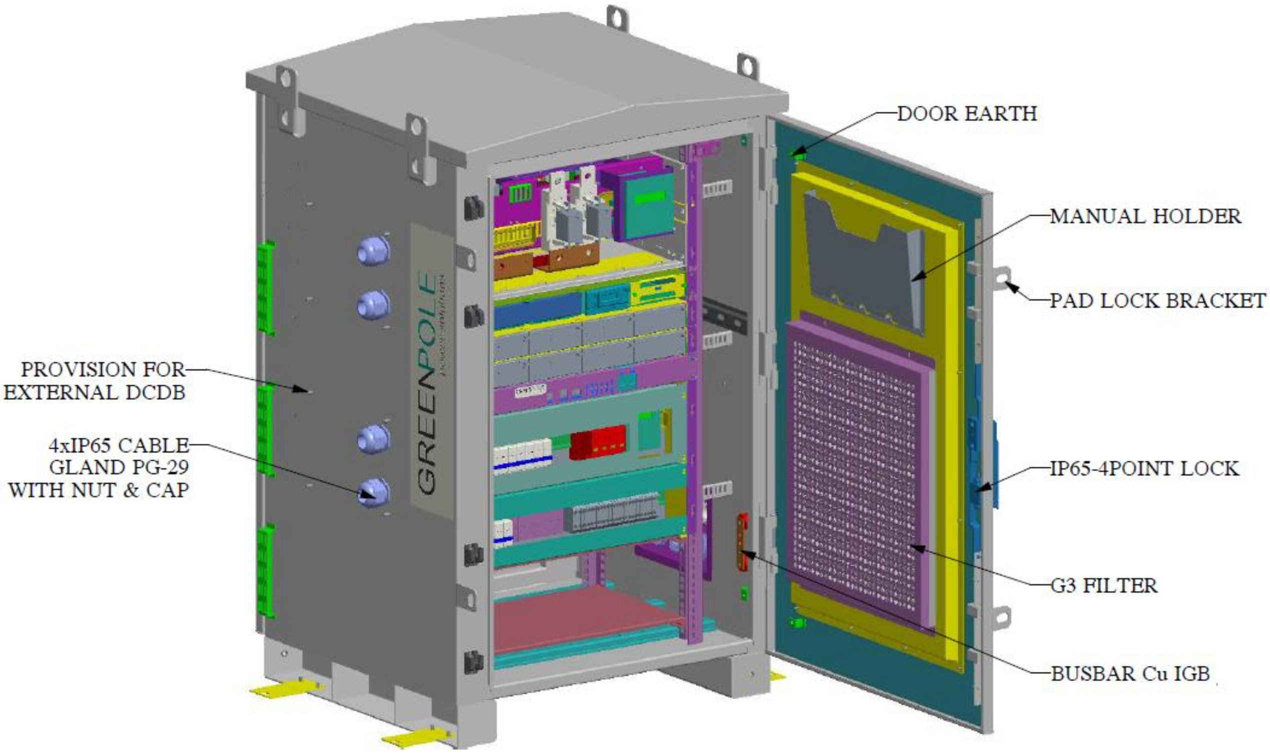
- When moving equipment by hand, wear protective gloves to prevent cutting of objects
- It is strictly prohibited to modify the structure of equipment and replace the position of plate parts without authorization, to replace components without authorization, and to drill holes on the cabinet without authorization. Non-conforming changes will affect the heat dissipation of equipment, electromagnetic shielding and other performance, self-drilling will also lead to metal chips into the cabinet, resulting in circuit board short circuits
- According to the specific situation of the site, select the rack placement position, the placement is based on the principle of considering the reasonable inlet and outlet lines of the equipment, leaving no less than 1m space between the front, back, right and left and the wall and other obstacles, and the installation height should be convenient for observation, monitoring and display, while considering the convenience of ventilation, heat dissipation and operation and maintenance, and shall be fixed on the frame or the floor with bolts, installation procedures of the system are as follows：

Refer following imaged for inner structure details of Power Cabinet.

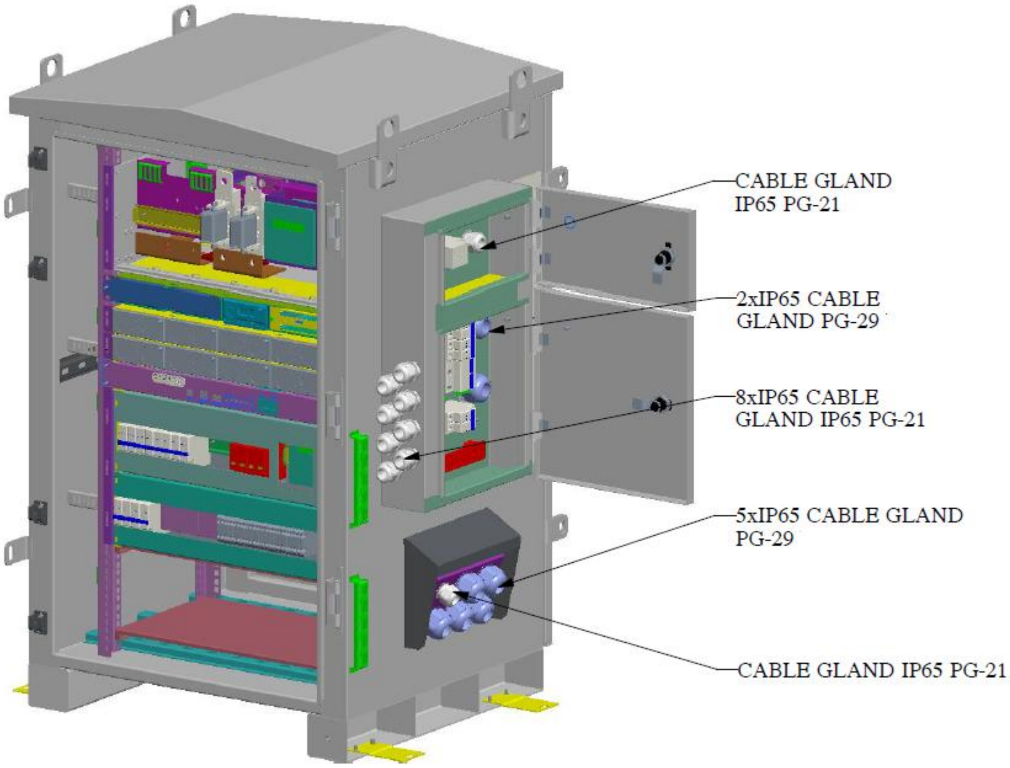




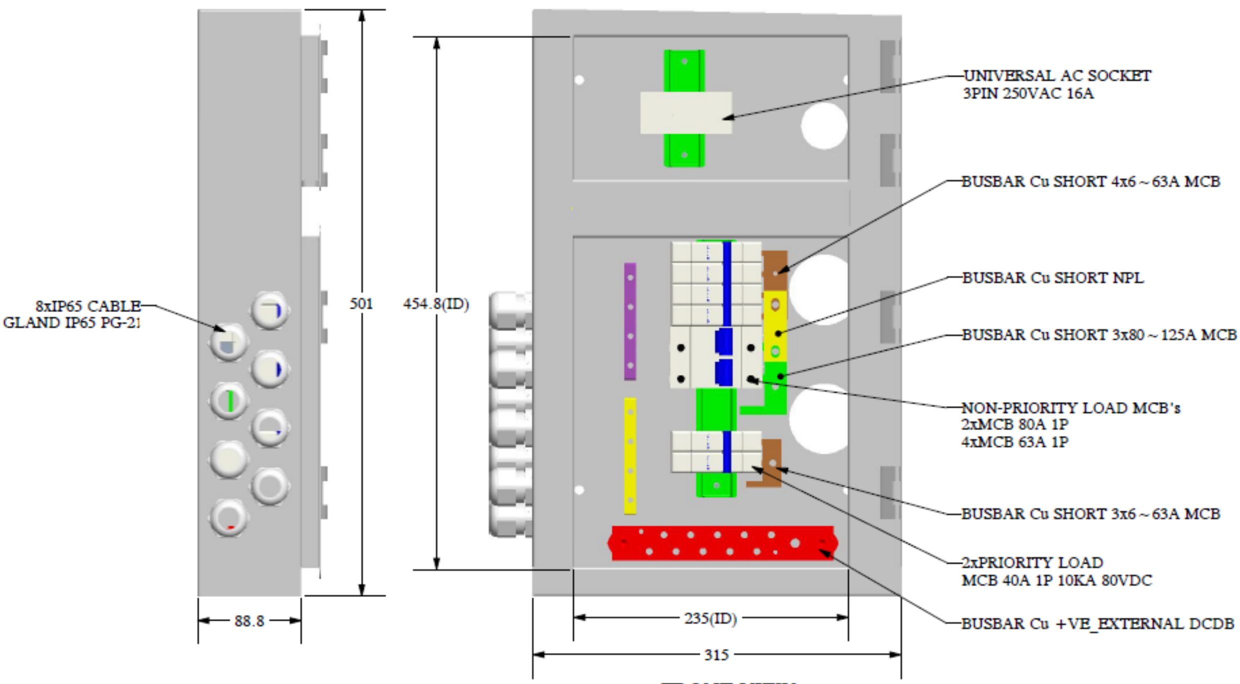
FRONT VIEW
FRONT DOOR HIDDEN



FRONT ISO VIEW



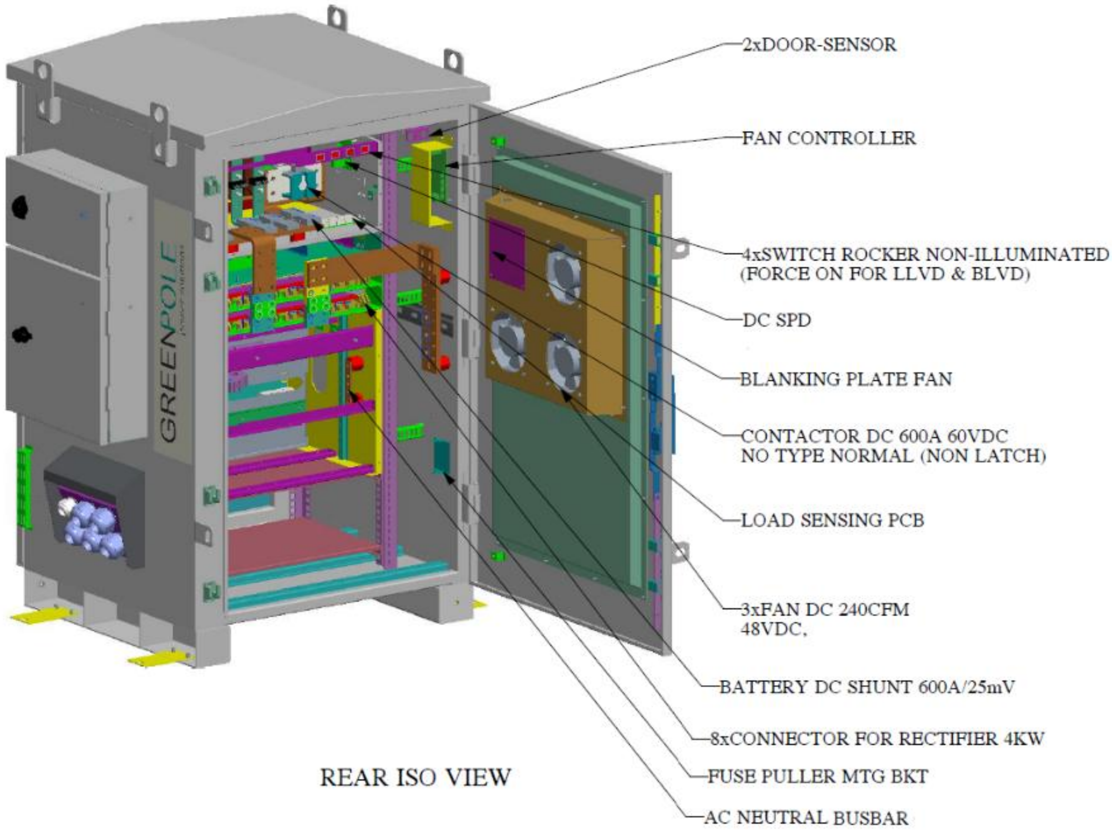
FRONT ISO VIEW
FRONT DOOR HIDDEN



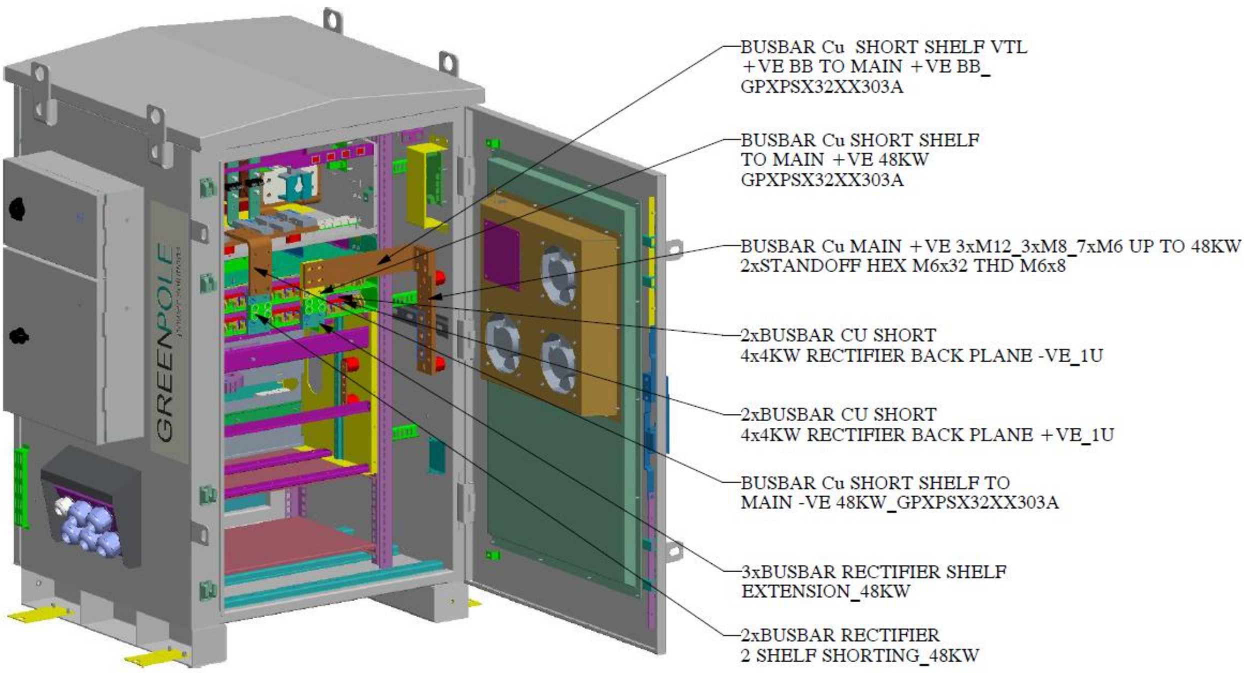
SIDE VIEW

FRONT VIEW
FRONT DOOR HIDDEN

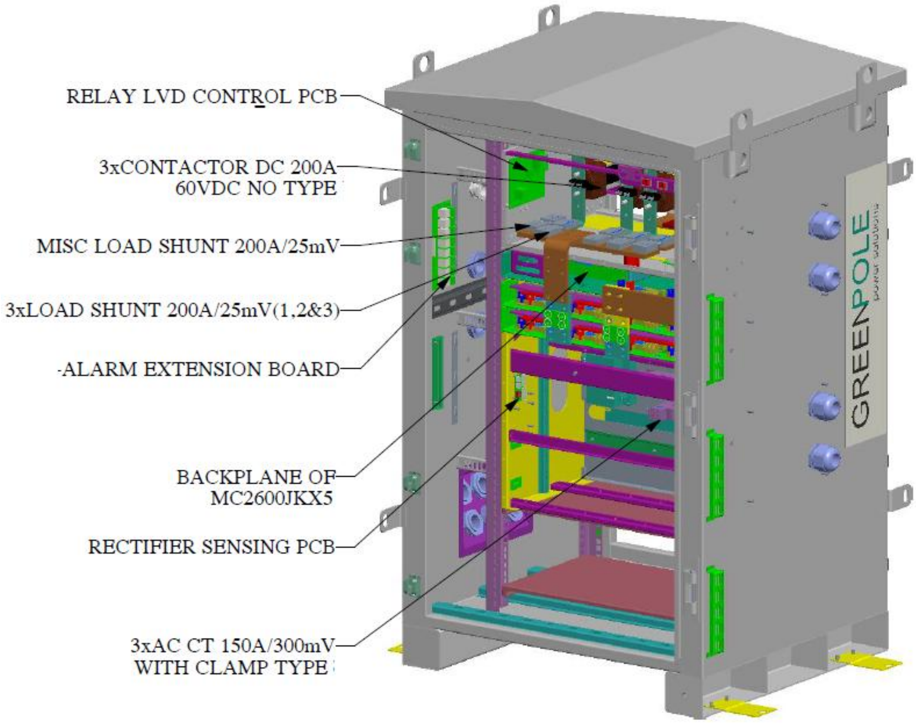
EXTERNAL DCDB ASSEMBLY



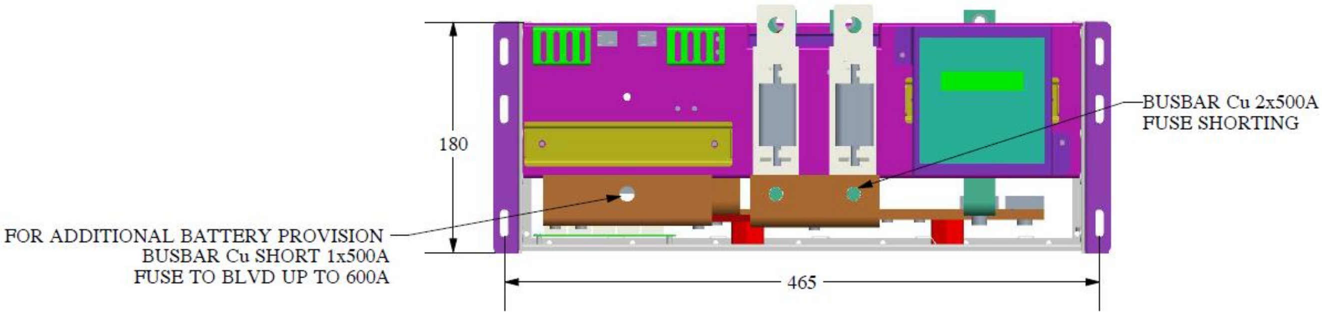
REAR ISO VIEW



REAR ISO VIEW



REAR ISO VIEW



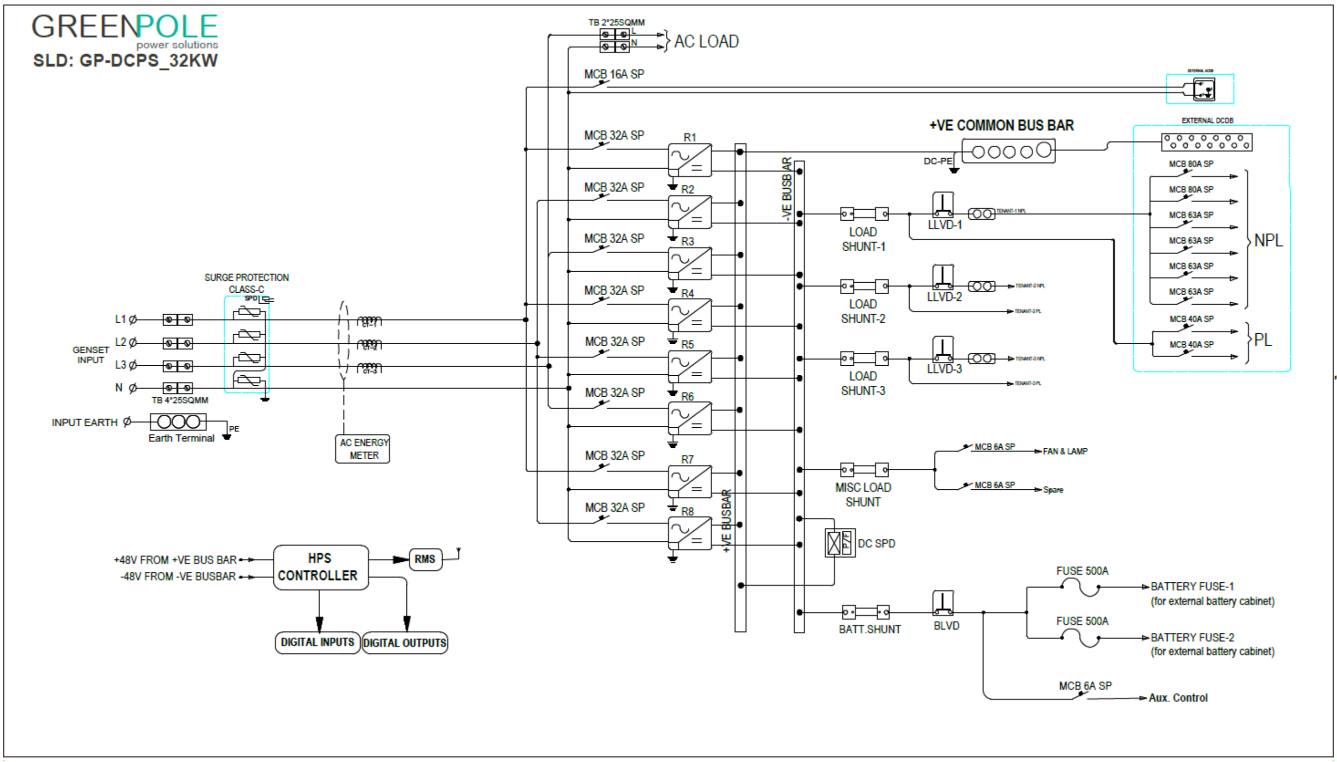
FRONT VIEW

10. System Configuration

10.1. System Component Detail

ITEM		CONFIGURATION
AC Input		AC Input: 4* 25Sqmm TB + 1*PE BUSBAR
AC distribution		AC Socket: 16A SP MCB
AC Input SPD		Class-C
Rectifier input		MCB 32A SP*8 Nos.
Rectifier Module		Type：GPR4000W* 8 Nos DC Rated Output Voltage：-42 to -58VDC
Controller		GP2600-B1
DC Distribution	Battery	2*500A FUSE
	DCDB	Tenant-1: NPL: 2*80A, 4*63A SP, PL: 2*40A SP MCB's, (External) Tenant-2: Bulk output Tenant-3: Bulk output MISC: 3*6A SP MCB for Fan/Lamp, Controller & spare
	DC SPD	Class C

10.2. System Single Line Diagram



Note:

- 1.To achieve better performance of the battery and the power system, the battery cabinet should be installed as close as possible to the power system to minimize the voltage cable drop.
- 2. Please ensure to use the same capacity and length being used for all the batteries connected in parallel using proper sized lug & Sleeves.

- 3.Whenever multiple battery cabinets are connecting it is advised to use same capacity and cable length across the cabinets.
- 4. Battery communication must be connected to Daisy chain method
- 5. Last battery of the existing cabinet communication connection to be connected to 1st battery of the new cabinet.
- 6. Communication terminal of last battery need to terminate with 120ohm resistance.
- 7.Battery communication details:
- a. For GP -V Battery: Pin 1 of RJ45 connector should be connected to B & Pin 2 of RJ45 connector should be connected to A.
- b. For GP -L Battery: Pin 1 of RJ45 connector should be connected to A & Pin 3 of RJ45 connector should be connected to B.

10.3. DC Power System Operation Principle

The DC power system converts the direct current generated by the single power sources i.e., AC input source. At the same time, the battery pack is equipped according to the user's demand, to realize the complementation with two sources. The priority of operation given first to AC input which means to deliver the load and battery charging if available. Below combinations will be perform by controller to provide continuous output:

1. During AC Input Available- Power will be delivered by the AC input to charge the battery and supply the load.
2. During AC Input Off- in this condition the load will be delivered by the battery

The controller can automatically inspect, monitor, and display the operating parameters of the near system, and can transmit all working data to the superior management platform through the RS485 interface.

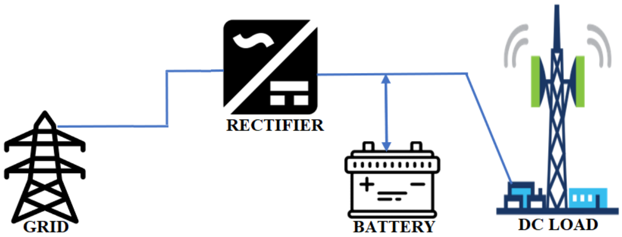


Figure 10.3.1 working principle block diagram of DC power system

10.4. Power Solution Features

Power System Controller main features as below：

- Safe and reliable Power, noise-free, pollution-free
- Modular design, hot plug, maintenance simple
- High voltage input, input isolation, safe and reliable
- Digital controller, soft switch
- Constant power output.
- Complete battery management automatically switching to Float mode, Boost mode and Deep-discharge protection.
- Regulating the charging voltage of the battery through temperature compensation, battery temperature monitoring
- 1000 history alarm records, 1000 data records

- Remote monitoring and controlling via SNMP
- Complete class - C lightning protection system

10.5. System Operation Environment

Temperature

Operation：–40℃～＋50℃ Storage：–40℃～＋70℃

Humidity

Operation：≤90%（No Condensation）; Storage：≤95%（No Condensation）

Atmospheric Pressure

Atmospheric Pressure (Altitude)：（70～106）KPa （Altitude：0～3000m）。

Vibration & Impact performance requirements

Vibration: the system can withstand frequency (10 ~ 55) Hz, the amplitude is 0.35mm, and the cycle time is 30min in the direction of each axis.

Impact: the rectifier module can withstand the peak acceleration of 150m/ s2 for a duration of 11ms.

The machine room shall be free from serious dust, explosive dangerous medium, harmful gas corroding metal and damaging insulation, conductive particles, and serious mold, and free from strong electromagnetic interference.

10.6. Electrical Cables Connection

Electrical cable connection refers below: Table 10.6.1

Table 10.6.1

Item	**Function description**
AC input cable	AC input: 16 sq.mm. 4core is recommended, the wires of Lines and N are red, Yellow, Blue & black respectively
Load and battery cable	Positive mounting holes are threaded holes. To choose cable and terminal according to load specification.
Note: the position of wire connection shall be in accordance with the system identification, screen printing and wiring schematic diagram. The suggestion of a cable compression terminal is only applicable to specific products.	

10.7. The Power Distribution Unit

AC/ DC input power distribution unit of power supply system and specifications are shown in below table.

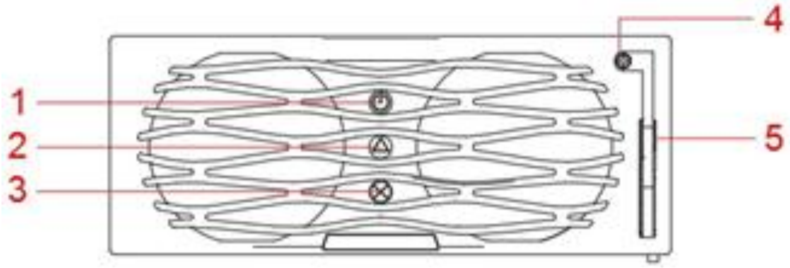
Table 10.7.1 Power distribution specifications

Item	Configurations
AC Input	3phase 4 wire with earth, 90 ~ 300VAC L-N
AC input termination	AC Input: 4*25SQMM TB
DC output power distribution	Battery Distribution: 2*500A FUSE Tenant-1: NPL: 2*80A, 4*63A SP & PL: 2*40A SP MCB’s(External)

10.8. Rectifier/Solar Module

The rectifier panel is as follows Figure 10.8.1.

Figure 10.8.1 Rectifier panel



1）Power indicator 2）Alarm indicator 3）Fault indicator 4）Locking latch 5）Handle

There are three indicators on the panel, which are used to reflect the operation status of the rectifier，see tab.10.8.1 for details.

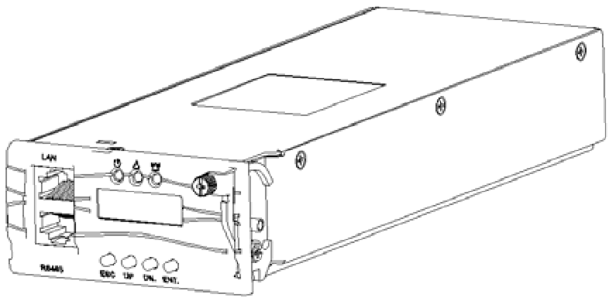
Table 10.8.1 Rectifier indicator description

Indicator	Color	Status	Reasons for abnormal status
Power indicator	Green	Normally on	The rectifier has an AC power input. The solar module has an DC power input.
		Off	Rectifier: Main supply fault (no AC input or OVP, UVP of AC input), non-output Solar module: PV supply fault (no PV input or OVP, UVP of PV input), non-output
Alarm indicator	Yellow	Normally on	Temperature alarm (OTP when the ambient temperature＞65℃)
			The rectifier is hibernating. (Indicator lighting, no alarm)
			The rectifier is current limiting
		Flickering	Communication failure
Fault indicator	Red	Normally on	Non-output caused by module inner reason, such as OVP, fan
		Normally off	The rectifier is running properly.

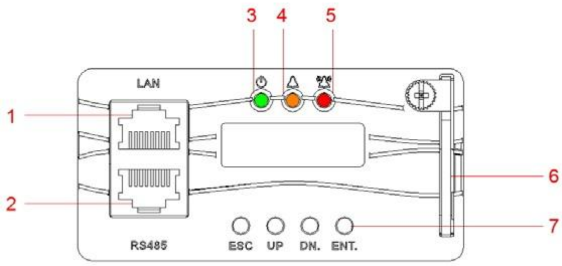
10.9. Monitoring Module

GP2600 monitor module is shown in figure 10.9.1

Figure 10.9.1 GP2600



The front panel of the GP2600 monitor module is shown in figure 10.9.2.



Front panel of the monitor module

（1）LAN port	（2）RS485 port	（3） Run indicator	（4）Minor alarm
（5）Major alarm indicator	（6）Handle	（7）Buttons	

Describes the buttons on the monitor panel is shown in table 10.9.1.

Table 10.9.1 Monitor button description

<table>
<tr><th>Button</th><th>Name</th><th colspan="2">Description</th></tr>
<tr><td>ESC</td><td>Return</td><td>Returns to the previous menu without saving the settings.</td><td rowspan="2">Press the ESC and ENT button at the same time within a short period of time can reset and restart the monitor.</td></tr>
<tr><td>ENT.</td><td>Confirm</td><td>Enter the main menu from the standby screen.
Enter a submenu from the main menu.
Saves the menu settings.</td></tr>
<tr><td>UP</td><td>Up</td><td>Turn to the previous menu or sets parameter values.
When setting parameter values, you can hold down this button to quickly adjust values.</td><td rowspan="2">When the parameter value is set by multiple string types, press up or down button to change each value.
After setting the value, press the confirm button to move the cursor back automatically</td></tr>
<tr><td>DN.</td><td>Down</td><td>Turn to the next menu or set parameter values.
When setting parameter values, you can hold down this button to quickly adjust values.</td></tr>
</table>

Describes the indicator on the monitor panel is shown in table 10.9.2.

Table 10.9.2 Indicator on the monitor panel

Type	Color	State	Instructions
Run indicator	Green	Normally on	Monitor is running properly
		Flickering	Monitor is running properly, but is not communicating
		Off	Monitoring is fault or has no DC input
Minor alarm indicator	Yellow	Normally on	The monitor generates a minor alarm.
		Off	The monitor does not generate any minor alarms.
Major alarm indicator	Red	Normally on	A major alarm is generated
		Off	No major alarm is generated

The monitor provides two communication ports（Reserve the RJ45 interface）The functional description is shown in the table

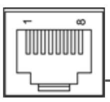


Table 10.9.3 RS485 Communication interface description table

Pin number	1	2	3	4	5	6	7	8
The signal	RS485+	-	RS485-	-	-	-	-	-

10.10. User Interface Board

The user interface board front panel as shown in figure 10.10.1：

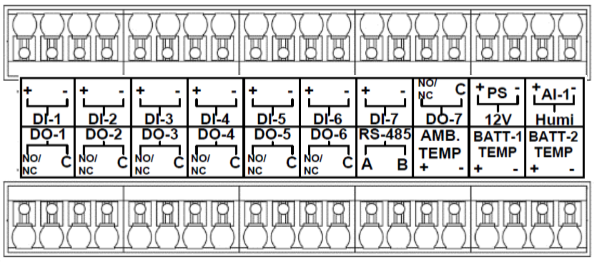


Figure10.10.1 user interface board front pane

User interface board contains 7 sets of DIs, respectively with DI1~DI7, Factory defaults in DI settings screen are shown in table 10.10.1

Table 10.10.1 Factory defaults in DI settings screen

Digital No.	Digital name	Normal Mode	Description
DI1	Digital1 Alarm	NO	Digital input 1 Alarm, Default settings:
DI2	Digital2 Alarm	NO	Digital input 2 Alarm, Default settings:
DI3	Digital3 Alarm	NO	Digital input 3 Alarm, Default settings.
DI4	Digital4 Alarm	NO	Digital input 4 Alarm, Default settings.
DI5	Digital5 Alarm	NO	Digital input 5 Alarm, Default settings. AC SPD Alarm Input
DI6	Digital6 Alarm	NO	Digital input 6 Alarm, Default settings. DI6/Mains Failure
DI7	Digital7 Alarm	NO	Digital input 7 Alarm, Default settings. DG Running Input

Wiring method:

1. With wire stripping pliers remove the dry contacts alarm signal lines insulation, wire core exposed length shall not be less than10 mm.
2. Use slot type screwdriver to press the terminal button at the top of the card spring to insert the cable terminal connection at the button of the mouth, loosen the button to complete the connection.
3. with handle gently pull cables, to ensure that the cable does not fall off.

10.11. Alarm Extension Board

Alarm extension board contains 7 set of DO's respectively DO1~ DO7, Factory defaults DO settings screen are shown in table 10.11.1

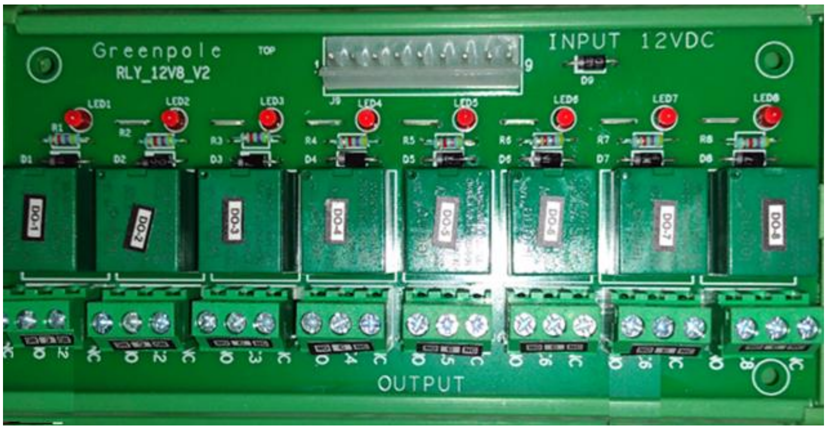


Table 10.11.1 Factory defaults in DO settings screen

Digital No.	Digital name	Normal Mode	Description
DO1	Relay output 1	NO	LOAD MCB TRIP / LLVD TRIP
DO2	Relay output 2	NO	AMBIENT/BATTERY TEMPERATURE HIGH/LOW
DO3	Relay output 3	NO	BATTERY DISCONNECT/ FUSE FAIL/BMS COM FAIL
DO4	Relay output 4	NO	AC INPUT VOLTAGE LOW/HIGH
DO5	Relay output 5	NO	SYSTEM DC VOLTAGE LOW/HIGH
DO6	Relay output 6	NO	SINGLE/MULTIPLE MODULE FAULT
DO7	Relay output 7	NO	DG REMOTE START SIGNAL

Wiring methods:

1. with wire stripping pliers remove the dry contacts alarm signal lines insulation or use crimp terminals to crimp the alarm cables, wire core exposed length shall not be less than 10 mm.
2. use screwdriver and connect the alarm cables to the respective DO relays C, NO, NC.
3. with handle gently pull cables, to ensure that the cable does not fall off.

10.12. Alarm & Communication termination details.

TB.No	DIGITAL INPUTS
1	DIGITAL INPUT-1 +VE
23	DIGITAL INPUT-1 COM
2	DIGITAL INPUT-2 +VE
24	DIGITAL INPUT-2 COM
3	DIGITAL INPUT-3 +VE
25	DIGITAL INPUT-3 COM
4	DIGITAL INPUT-4 +VE
26	DIGITAL INPUT-4 COM
5	DIGITAL INPUT-5 +VE
27	DIGITAL INPUT-5 COM
6	DIGITAL INPUT-6 +VE
28	DIGITAL INPUT-6 COM
7	DIGITAL INPUT-7 +VE
29	DIGITAL INPUT-7 COM
8	DIGITAL INPUT-8 +VE
30	DIGITAL INPUT-8 COM
9	DIGITAL INPUT-9 +VE
31	DIGITAL INPUT-9 COM
10	DIGITAL INPUT-10 +VE
32	DIGITAL INPUT-10 COM
11	DIGITAL INPUT-11 +VE
33	DIGITAL INPUT-11 COM
12	DIGITAL INPUT-12 +VE
34	DIGITAL INPUT-12 COM
13	DIGITAL INPUT-13 +VE
35	DIGITAL INPUT-13 COM
14	DIGITAL INPUT-14 +VE
36	DIGITAL INPUT-14 COM

TB.No	DIGITAL OUTPUT
15	DIGITAL OUTPUT-1 COM
37	DIGITAL OUTPUT-1 NO/NC
16	DIGITAL OUTPUT-2 COM
38	DIGITAL OUTPUT-2 NO/NC
17	DIGITAL OUTPUT-3 COM
39	DIGITAL OUTPUT-3 NO/NC
18	DIGITAL OUTPUT-4 COM
40	DIGITAL OUTPUT-4 NO/NC
19	DIGITAL OUTPUT-5 COM
41	DIGITAL OUTPUT-5 NO/NC
20	DIGITAL OUTPUT-6 COM
42	DIGITAL OUTPUT-6 NO/NC
21	DIGITAL OUTPUT-7 COM
43	DIGITAL OUTPUT-7 NO/NC
22	DIGITAL OUTPUT-8 COM
44	DIGITAL OUTPUT-8 NO/NC

TB.No	COMMUNICATION
1	RS 485-1 A (+VE)
6	RS 485-1 B(-VE)
2	RS 485-2 A (+VE)
7	RS 485-2 B(-VE)
3	To Solar CT 5V Input
8	From Solar CT Output
4	To Solar CT GND
9	Not Used
5	DG ON Sensing_Line (R-Ph)
10	DG ON Sensing_Neutral

11. Main LCD Screens

In this chapter, several display screens are frequently mentioned; this section introduces the display content on and the approach to entering each screen.

11.1. System Information Screen

When the monitoring module is powered on, information display screen appears as Fig 11.1.1 shows.

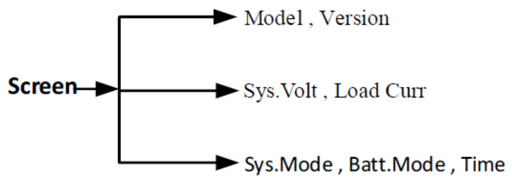


Fig 11.1.1 Information display screen

The system information screen shows the main information, including system voltage, load current, system control mode, battery charging state and date. System control mode including of Auto and Manual, Battery charging state including of float charge, boost charge, temperature compensation and test, etc.

1. The first screen system voltage and load current will be displayed after monitor module power-on.

Sys. Volt : 54V

Load Curr : 10A

2. At the Main Menu screen, press 'ESC' button to return to the first system information screen.
3. If no operation is conducted on the monitoring module button for 2minutes, the LCD will return to the first system information screen.
4. Press 'ESC' button on the first screen, the model and software version will be displayed.

Model: MC2600

SW Ver: X.XXXXXX

5. Press 'DN.' button on the first screen, system time setting screen will be displayed, set the time by 'UP' and 'DN.' button.

Auto Float

2020-01-09/10:30:00

11.2. Main Menu Screen

The main menu is the top menu of the monitoring module. All settings, controls and status of rectifier module information and alarm information are achieved through the submenus of the main menu. Press the ESC button on any submenu of the main menu to return to the main menu screen level by level. Fig. 11.2.1 shows the main menu.

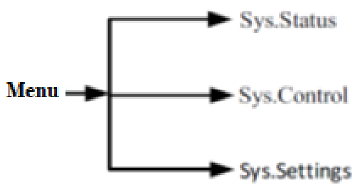


Fig. 11.2.1 Main menu screen

11.3. Information Inquiry Screen

Information inquiry screen is submenu of main menu. Information inquiry screen refers to Fig 11.3.1.

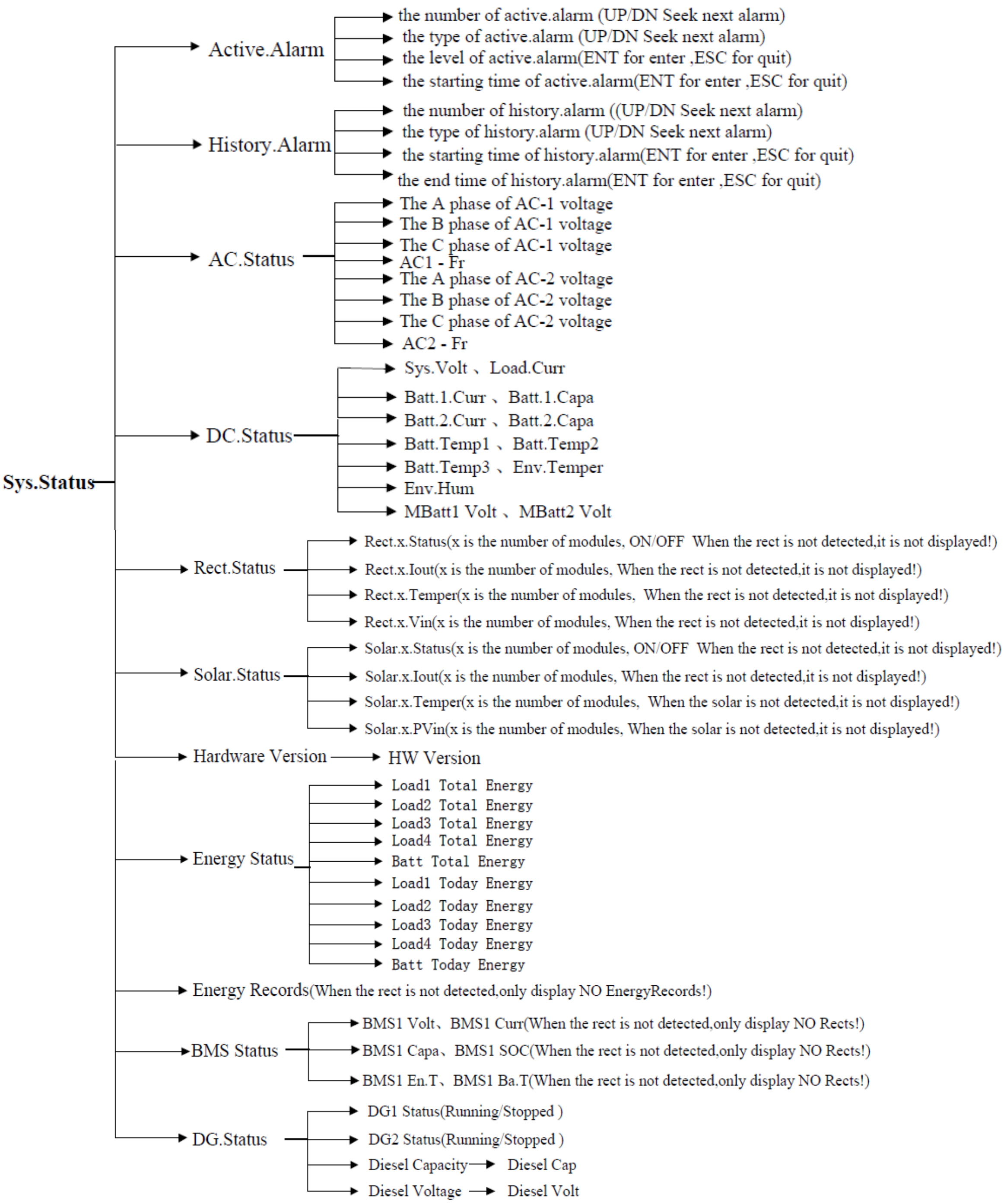


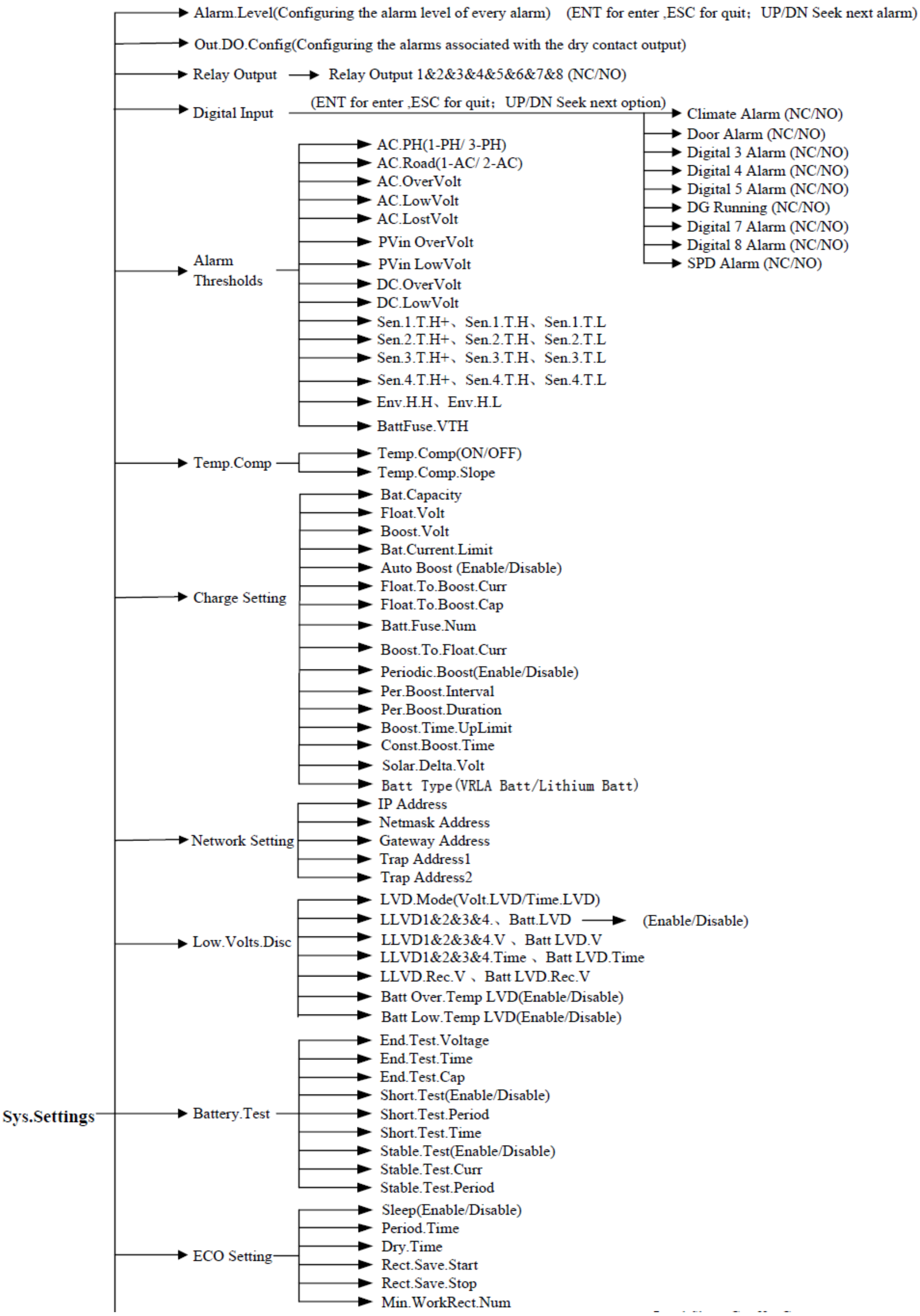
Fig 11.3.1 Information inquiry screen

1. Press ‘UP’ and ‘DN.’ button to select ‘Information inquiry’, press ‘ENT’ button to confirm, then goes into Information inquiry screen.
MC2600 Controller User Manual
2. All displayed data in ‘Information Inquiry screen’ submenu just can be read but can’t be edited.
3. Press ‘ESC’ button on the Information inquiry submenu, then go back to Information inquiry screen step by step.

4. When the user check the current alarm or history alarm, the alarm name will be displayed directly after entering the submenu. Press ‘ENT’ button to enter and ‘ESC’ button to back if user want check alarm time. Press ‘UP’ or ‘DN’ button to check next alarm.

11.4. Setting Screen

The setting screen is a Level 1 submenu screen of the main menu. There are also several submenu screens for setting all parameters of the power system. The setting screen refers to Fig. 11.4.1.



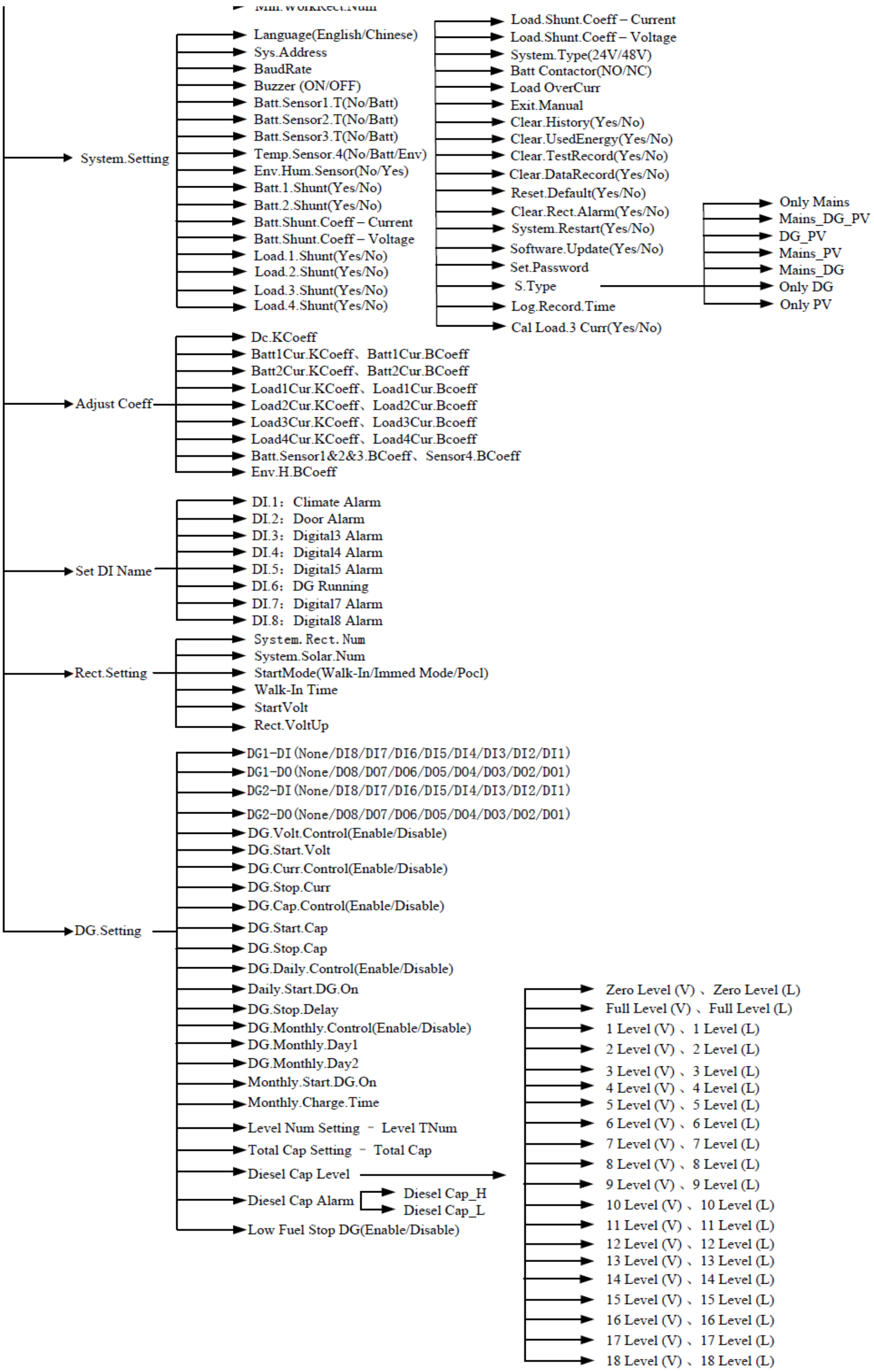


Fig 11.4.1 Information inquiry screen

Notice: Press 'ENT' button to enter 'Settings', press 'UP' and 'DN' button to choose 'Out DO Config', and then press 'ENT' button to enter it. Now it display the first alarm setting option, you can press 'ENT' button to modify it and press 'UP' or "DN' to next option.

After you choose your option, you need to press 'ENT' button to make the set success. If you do not need to set the current alarm setting option, you can press 'DN' to select next alarm setting. When you are finished, press 'ESC' to return to the previous menu.

11.5. Alarm Level

Alarm setting screen is a submenu of parameter setting menu, mainly use for setting alarms level of the system.

Move the cursor to the option need be changed by press 'UP' and 'DN.' button, then press 'ENT button into modification mode, choose the related content by press 'UP' and 'DN.' button. Press 'ENT' button to confirm at last.

Monitoring module divided alarms type into three levels: Major alarm, Minor alarm, no alarm.

Major alarm: This alarm affects power system working performance severely. Whenever it appears, user must take method to deal with it. Red alarm LED lighting with voice alarm.

Minor alarm: Power system can maintain normal DC output temporally when this alarm appears. When it happens during duty time, method must be taken to deal with it. If not, deal with it when workers on duty. Yellow alarm LED lighting when alarm appears.

The default alarm settings refer to Table 10.10.1. & 10.11.1.

11.6. Alarm Thresholds

Alarm threshold is submenu of parameter setting menu, mainly used by operator for setting each alarm in system.

Press the UP or DN button to select the parameter to be set, and press the ENT button to confirm the selection, then, press the UP or DN button again to select the parameter value, and press the ENT button to confirm and save the value. Alarm threshold parameter settings refer to Table 11.6.1.

Table 11.6.1 Alarm threshold parameter setting

Parameter name	Setting range	Default value	Setting description
AC PH	1-PH/3-PH	3-PH	Set according to system actual configuration, choose single phrase or three phrases
AC Road (Source)	1-AC/2-AC	1-AC	Set according to system actual configuration, choose only one AC input or two exchanges.
AC Over Volt	AC Low Volt～500V	300V	AC input voltage is higher than setting value, AC over voltage alarm appears. AC overvoltage value is higher than AC low voltage value.
AC Low Volt	AC Lost Volt～AC Over Volt	90V	AC input voltage is lower than setting value, AC over voltage alarm appears. AC low voltage value is lower than AC over voltage value, and higher than AC lost voltage value.
AC Lost Volt	50V～AC Low Volt	80V	AC input phrase voltage lower than 'phrase alarm' setting value. AC phrase fail alarm appears. AC lost voltage value is lower than AC low voltage value
PVin Over Volt	80V～600V	440V	MPPT input DC voltage higher than setting value. PV in over voltage alarm appears. PV Input over voltage must be higher than PV Input low voltage.

PVin Low Volt	80V～600V	100V	MPPT input DC voltage lower than setting value. PV in low voltage alarm appears. PV Input low voltage must be lower than PV Input over voltage.

11.7. Temperature Compensation

Temperature compensation screen is submenu of parameter setting, mainly used for operator to set temperature compensation parameter of battery management.

Press the UP or DN button to select the parameter to be set, and press the ENT button to confirm the selection, then, press the UP or DN button again to select the parameter value, and press the ENT button to confirm and save the value.

Alarm threshold parameter setting description please refer to Table 11.7.1

Table 11.7.1 Temperature compensation parameter setting

Parameter name	Setting range	Default value	Setting description
Temp. Comp	ON, OFF	OFF	ON: Open temperature compensation function OFF: Close temperature compensation function Notice: 1. The temperature compensation function fail when all battery sensor is set as ‘no’; 2. The temperature compensation function is invalid when battery type is set to lithium battery mode.
Temp. Comp. Slope	0～500mV/℃	72mV/℃	Float charge voltage decrease value = (Battery temperature measuring value- 25) × Temperature compensation coefficient When rectifier communication fails or system over/under voltage, temperature compensation function will be invalid. When anyone battery temperature sensor is fault or disconnect, temperature compensation function will be invalid. When there are multiple temperature sensors, select the lowest temperature for temperature compensation

11.8. Charge Setting

Press the UP or DN button to select the parameter to be set and press the ENT button to confirm the selection; then, press the UP or DN button again to select the parameter value, and press the ENT button to confirm and save the value. The charging management parameter value description is listed in Table 11.8.1.

Table 11.8.1 Battery charging parameter setting description

Parameter name	Setting range	Default value	Setting description
Bat Capacity	12～5000Ah	600Ah	Bat Capacity is known as the nominal capacity of the battery. You should set this parameter according to the actual battery configuration. If you have multiple batteries in parallel, It should be set to the sum of the battery rated capacity of the installation.

Float Volt	42V～Boost Volts	52.5V	In the FC state, all rectifiers output voltage according to the set float voltage. The float voltage must be lower than boost voltage.
Boost Volt	Float Volts～58V	53.5V	In the BC state, all rectifiers output voltage according to the set boost voltage. The boost voltage must be higher than the float voltage.
Bat.Current Limit	0.1～0.6C10	0.2C10	This is the maximum charging current that should be allowed into the battery at any time, as regards to the nominal capacity of the battery. For example, a value of 0.2C10 means that the charging current is limited to 20% of the battery's nominal capacity. Notice: C10 means the nominal capacity of the battery.
Auto Boost	Enable, Disable	Disable	Enable: meet boost condition, transfer to boost charge automatically. Disable: transfer to boost charge isn't allowed. Notice: The boost function is invalid when battery type is set to lithium battery mode.
Float to Boost Curr	0.040～0.080C10	0.05C10	The monitoring module will control the system enter the BC state when the battery capacity decreases to the value of To Boost Capacity, or when the charge current reaches the To Boost Current. The charge voltage will be the Boost
Float To Boost Cap	10％～99％	80％	
Batt.Fuse. Num	1-4	1	Set the number of battery fuses in the system.
Boost To Float Curr	0.002～0.02C10	0.02C10	'Boost To Float Current' is also known as constant boost current. When the system is on boost charge status, if charge current is lower than 'Boost To Float Current' setting values， system will enters constant boost charge status. When the system is on constant boost charge status, if charge timer is more than 'Const Boost Time' setting values, system enters the float charge status.
Const Boost Time	5～1440min	180min	
Periodic Boost	Enable, Disable	Disable	Enable: Use this function Disable: Do not use this function Notice: The boost function is invalid when battery type is set to lithium battery mode.
Per.Boost Interval	48～8760 hours	2400hours	To set cyclic boost charging. 'Per.Boost Interval' means interval time between twice timing boost charge. Battery charging voltage is 'Boost charge voltage' Setting value. Charge time is 'Per.Boost Duration' setting value.
Per.Boost Duration	30～2880min	240min	
Boost Time UpLimit	60～2880min	1080min	During boost charge, monitor module will force system turns to float charge to secure system when boost

			charge time reaches to the value of ‘Boost charge protection time’
Solar Delta Volt	0～2V	1V	Set the threshold of voltage difference between solar module and rectifier module. Make the solar module work as a priority. (Suitable for mixed system of rectifier module and solar module)
Batt Type	VRLA Batt Lithium Batt	Lithium Batt	Set the type of battery in the system. Two different battery current limiting management strategies.

The BC/FC switchover diagram is shown in Fig. 11.8.1

FC time longer than "Scheduled BC Interval"

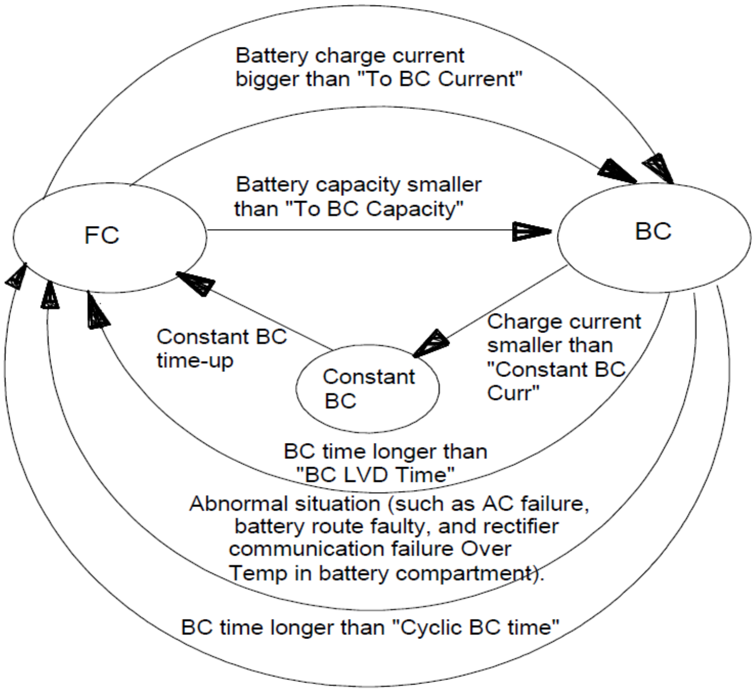


Fig 11.8.1 The BC/FC switchover diagram

11.9. Load LVD and Battery LVD

Load LVD and Battery LVD is submenu of parameter setting menu, mainly used for battery disconnect protection and load disconnect protection of battery management.

Press ‘UP’ or ‘DN.’ button to select parameter which need be set, press ‘ENT’ button to confirm. Then press ‘UP’ and ‘DN.’ button to select parameter again, press ‘ENT’ button to confirm and save at last.

Load disconnect means AC power-off, battery supply power and system cut-off minor important load automatically to prolonged power provided time of major important load

Battery protection means AC power-off, battery provide power, system cut-off battery automatically to avoid the situation that battery life be shorten caused by secondary cell discharge too much.

Set alarm threshold parameter description table shown in Table 11.9.1.

Table 11.9.1 LVD parameter setting description

Parameter name	Setting range	Default value	Setting description
LVD Mode	Volt LVD, Time LVD	Volt LVD	Volt LVD: LVD according to voltage. Time LVD: LVD according to time.
LLVD1	Enable, Disable	Enable	Enable: Can use load.1~4 disconnect function Disable: Cannot use load.1~4 disconnect function
LLVD2	Enable, Disable	Disable	
LLVD3	Enable, Disable	Disable	
LLVD4	Enable, Disable	Disable	
LLVD1.V~LLVD4.V	BLVD.V～56V	48V	When system voltage decreases to this setting voltage, load.1~4 contactor can be disconnected. Load.1~4 LVD voltage should be higher than Batt LVD voltage
LLVD.Rec.V	37V～58V	49.5V	When system voltage increases to this setting voltage, load.1~4 contactor can be connected. LLVD1~4. Rec. V Should be higher than LVD1~4. V. Notice: This is the forced power recovery voltage, not associated with the ac or pv input voltage. When the ac or pv input voltage is normal, the power on does not refer to this voltage.
LLVD1.Time~ LLVD4.Time	5～3600min	300min	When mains failure time exceeds this setting time, load.1~4 contactor can be disconnected. LLVD1~4 time should be lower than battery LVD time.
Batt LVD	Enable, Disable	Enable	Enable: Can use battery LVD function Disable: Cannot use battery LVD function
Batt LVD.V	35V～58V	46.2V	When battery voltage decreases to this setting voltage, BLVD appears.
BattLVD.Rec.V	35V～58V	47.5V	When battery voltage increases to this setting voltage, battery contactor can be connected. Recovery voltage should be higher than disconnected voltage. Notice: This is the forced power recovery voltage, not associated with the ac or pv input voltage. When the ac or pv input voltage is normal, the power on does not refer to this voltage.
BattLVD.Time	5～3600min	360min	When mains failure time exceeds this setting time, battery contactor can be disconnected.
Batt Over.Temp LVD	Enable, Disable	Disable	When anyone battery temperature is higher than over temperature alarm value (Default: 60 ℃), battery contactor can be disconnected. This function can be enabled or disabled by LCD or client.

Batt Low.Temp LVD	Enable, Disable	Disable	When anyone battery temperature is lower than low temperature alarm value (Default: -33℃), battery contactor can be disconnected. This function can be enabled or disabled by LCD or client.

11.10. Battery Test

The battery test process allows for the battery to be discharged on-line using the system load. There are three types of battery test:

- Manual start and stop stable tester short test.
- Automatic table test, if enabled.
- Automatic short test, if enabled.

Battery test screen is submenu of parameter setting screen, mainly used for operator to set battery test of system battery management.

Press ‘UP’ or ‘DN.’ button to select parameter which need be set, press ‘ENT’ button to confirm. Then press ‘UP’ or ‘DN’ button to select parameter again, press ’ENT’ button to confirm and save at last.

Parameter setting range, default value and setting description of battery refer to Table 11.10.1.

Table 11.10.1 Battery testing parameter setting description (default settings considering VRLA battery)

Parameter name	Setting range	Default value	Setting description
End Test Voltage	43.1V～57.9V	48.4V	In the process of battery test, if meet one of the following three conditions, exit the battery test and turn to float charge: a) Battery voltage reaches to ‘End Test Voltage’ b) Battery discharge time reaches to ‘End Test Time c) Battery residual capacity reaches to ‘End Test Cap’
End Test Time	5～1440min	300min	
End Test Cap	1％～95％	40％	
Short Test	Enable, Disable	Disable	Enable: Can use fast test function Disable: Can’t use fast test function
Parameter name	Setting range	Default value	Setting description
Short Test Period	24～8760H	720H	A short test is a short duration battery discharge test. Short Test Period is the interval, in hours, between the shortest cycles. Short Test Time is the duration, in minutes, of each short battery test. When start the short test, the rectifier modules will be turned down to a voltage just below ‘End Test Voltage’, then battery began to discharge. In the short test time, if the short test fault alarm appears, battery short test is failed. If the short test fault alarm does not appear, battery short testis pass. If there is only one battery current in the power system, short test function is invalid. If there are just two battery current in the power system, short test function is valid.
Short Test Time	1～60min	5min	

Stable Test	Enable, Disable	Disable	Enable: Can use Stable Test function Disable: Cannot use Stable Test function
Stable TestCurr	0.1C10～0.9 C10	0.2C10	When start the stable test, the battery discharge current is limited to ‘Stable Test Curr’, via adjust the current limit value of rectifier modules. Stable test means battery discharge according to constant current. Set current value according to ‘Stable Test Curr’. Stable test mode is recommended in only one battery current system if you want to test battery. Stable test current can be set to 1.0C If you want to keep battery discharge current to full capacity.
Stable Test Period	24～8760H	4320H	Stable Test Period is known as cyclic stable test period. This is the interval, in hours, between the stable test cycles. In the process of battery stable test, if test time or Ah is obtained batteries are good, the battery test fault alarm does not appear, battery testis pass; If end voltage is reached first batteries are bad, the battery test fault alarm appears, battery test is fail. In other words, when the battery test time has reached 'End Test Time’, or the battery capacity has reached ‘End Test Cap', but the battery voltage has not yet reached 'End Test Voltage', the battery test fault alarm does not appear. Battery test is pass. When the battery voltage has reached 'End Test Voltage', but the battery test time has not yet reached 'End Test Time’, and the battery capacity has not reached 'End Test Cap', the battery test fault alarm appears. The battery test fault alarm disappears after 5 minutes and become a historical alarm. Notice: This type of battery test is designed to give an indication only of battery state of health.

Test function operating principles shows in bellowing Fig 11.10.1:

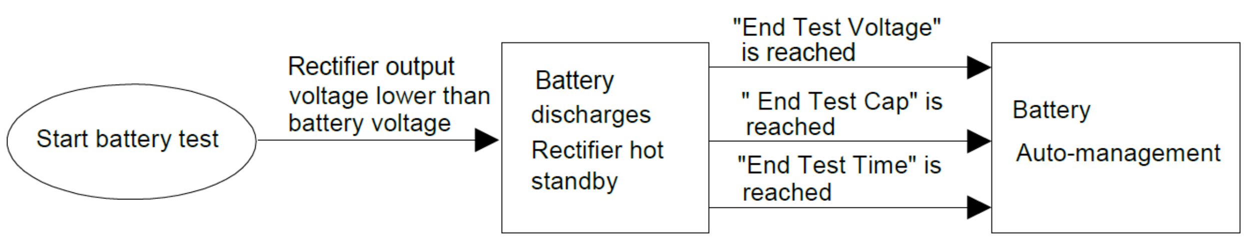


Fig 11.10.1Testing function operating principles

During the battery test, if meet the following anyone condition, exit the battery test and turn to float charge:

- Mains Failure and PV input fail
- BLVD, Battery fuse alarm, Rectifier communication fail,
- Battery temperature low/high/ higher
- Battery discharge time reaches to ‘Short Test Time’, if short test enabled.
- Battery voltage reaches to ‘End Test Voltage’.
- Battery discharge time reaches to ‘End Test Time’
- Battery residual capacity reaches to ‘End Test Cap’

11.11. ECO Setting

ECO setting screen is submenu of parameter setting screen, mainly use for setting system ECO parameter to save energy.

1. Operating principles
   In the ECO mode, the monitoring module controls some rectifier modules to turn off and makes the rectifier modules powered on bear all loads. Each rectifier module powered on works at the optimal efficiency point whenever possible, to improve the utilization rate of the rectifier module and reduce energy consumption. For rectifier modules having entered the off state, the monitoring module will power them on after a certain period (i.e., the set value of “Period Time” shown in Table. 2-13); after these rectifier modules run for some time, the monitoring module will control the rectifier modules having worked for a long period to turn off again. The two states switch in cycles, ensuring that the rectifier modules in the system start working closely. In case of change in battery current or load current in the system, the monitoring module will control some working rectifier modules to turn off according to the actual situation or control some rectifier modules in off state to turn on and start working. Under any circumstances, the system will ensure that at least one rectifier module works after turned on.
2. Preconditions
   The system can run in the ECO mode only when the system is equipped with battery and the load current has no instant oscillation (i.e., the parameter of “Sleep” is set to “Enable”.)
3. Advantages
   The ECO mode enables rectifier modules to work at the optimal efficiency point, therefore can save power for the system.
   It can balance the working time of rectifier modules in the system and prolong the service life of rectifier modules.
   Rectifier modules in off state can avoid damage caused by AC impact and reduce failures caused by lightning stroke, etc.
4. Handling of abnormalities
   In case of system voltage fault (DC under voltage), all rectifier modules will turn on.
   In case of rectifier module alarms (e.g., communication interruption), all rectifier modules will turn on.
   In case of no controller in location interruption between rectifier modules and controller, all rectifier modules will turn on automatically.

   ECO parameters setting description refers to Table 11.11.1.

Table 11.11.1 ECO parameter setting description

Parameter name	Setting range	Default value	Setting description
Sleep	Enable, Disable	Disable	Enable: Can use fast test function. Disable: Can't use fast test function Notice: Only can be set as 'Enable' if system has battery and no large load current shock appears.
Period Time	1～8760h	1h	Period Time is also known as cycle activation time. In order to synchronize the life of all the rectifier modules, it is necessary to operate the module with a long sleep time at a certain time interval while allowing the module with a long working time to sleep. This time interval is cycle activation time.
Dry time	10~240min	120min	After exiting the last sleep mode, the work time of rectifiers on the no sleep mode. It is allowed to enter sleep mode again when more than dry time.
Rect Save Start	30%～90%	60%	Rect Save Start is also known as rectifier best operating point. At the best operating-point, the rectifier module operates at a relatively high efficiency and is subject to more suitable stresses. For energy saving, the power supply system should try to make the rectifier module load / rated capacity ratio as close as possible to the best operating-point. Notice: Load / Rated capacity ratio = Rectifier real output current / Rectifier rated current * 100%.
Rect Save Stop	30%～100%	90%	Rect Save Stop is also known as system energy-saving point. If load / rated capacity ratio more than system energy-saving point, the power supply system exits energy-saving mode.
Min. Work Rect Num	1～20	3	The minimum number of rectifiers working.

11.12. System Setting

The system setting screen is submenu screen under the main menu, mainly used for operator to set parameter according to system hardware configuration, then to reduce part of system functions but make sure system can running well at the same time.

System parameter setting description refers to Table 11.12.1:

Table 11.12.1 System parameter setting description

Parameter name	Setting range	Default value	Setting description
Language	Chinese, English	English	Set according to user need.
Sys. Address	0～255	32	Controller communication address.
Baud Rate	2400bps, 4800bps,	9600bps	Serial communication baud rate for the RS485 port on front panel. Set according to user need.

	9600bps, 19200bps, 38400bps		
Buzzer	ON, OFF	ON	ON: open voice alarm function OFF: close voice alarm function
Batt.Sensor1.T	Batt, No	Batt	Batt: Enable sensor 1 for battery temperature. No: Disable sensor 1 for battery temperature. Set according to user need. Temperature compensation function can be used for VRLA battery if the power system has battery temperature sensor 1.
Batt.Sensor2.T	Batt, No	No	Batt: Enable sensor 2 for battery temperature. No: Disable sensor 2 for battery temperature. Set according to user need. Temperature compensation function can be used for VRLA battery if the power system has battery temperature sensor 2.
Batt.Sensor3.T	Batt, No	No	Batt: Enable sensor 3 for battery temperature. No: Disable sensor 3 for battery temperature. Set according to user need. Temperature compensation function can be used for VRLA battery if the power system has battery temperature sensor 3.
Temp.Sensor.4	Batt, Env, No	Env	Set according to user need. It can be configured for battery temperature or ambient temperature.
Env.Hum.Sensor	Yes, No	No	Set according to user need.
Batt.1 Shunt	Yes, No	Yes	Set according to actual power system. Battery shunt should be set as 'No' if system doesn't sample via shunt.
Batt.2 Shunt	Yes, No	No	
Batt.Shunt Coefficient - Current	1～2000A	300A	Set according to actual power system.
Batt.Shunt Coefficient - Voltage	1～500mV	25mV	
Load 1 Shunt	Yes, No	Yes	Set according to actual power system. If all load shunt is set to 'No', the total load current is equal to the sum of module currents minus the sum of battery currents.
Load 2 Shunt	Yes, No	No	
Load 3 Shunt	Yes, No	No	
Load 4 Shunt	Yes, No	No	
Load Shunt Coefficient - Current	1～2000A	300A	Set according to actual power system. Notice: These two parameters will be invalid if all load shunt is configured as No.

Load Shunt Coefficient - Voltage	1～500mV	25mV	
System Type	24V,48V	48V	Set according to actual situation.
Batt Contactor	NC, NO	NO	Battery contactor type is normal open.
Load OverCurr	80%~120%	100%	System overload alarm percentage. For example, if load over current alarm value is 80% in 200A system, when load current more than 160A, load over current alarm will appear. If load over current alarm value is 100% in 200A system, when load current more than 200A, load over current alarm will appear.
Exit Manual	None, 30min, 1hour, 2hour, 4hour	30min	Manual to auto mode timer can be configurable. None means that manual mode will not be switched to automatic mode unless there is dc under voltage alarm.
Clear History	Yes, No	No	Used for clearing all history alarm record.
Clear Used Energy	Yes, No	No	Used for clearing all used energy record.
Clear Test Record	Yes, No	No	Used for clearing all battery test record.
Clear Data Record	Yes, No	No	Used for clearing all history data record.
Reset Default	Yes, No	No	Used for resetting system factory default setting.
Clear Rect Alarm	Yes, No	No	Used for clearing all rectifiers alarm.
System Restart	Yes, No	No	Used for restarting system controller.
Software Update	Yes, No	No	Used for software updating, unprofessional operator is not allowed to do this.
Set password	No more than 6 digits or characters	2	If you need to set parameters and controls, you must enter a password to have permission. If there is no button operation for more than 2 minutes, you need to re-enter the password.
S.Type	Only Mains, Only PV, Only DG, Mains_DG, Mains_PV, DG_PV, Mains_DG_PV	Mains_DG	DC Power system energy input sources type. Set according to actual power system. Notice: Mains means AC input voltage; PV means solar panels; DG means diesel generators.
Log Record Time	1~1440min	5min	History data logs record period Time.
Cal Load 3 Curr	Yes, No	No	Calculate the load 3 current

11.13. Set Digital Input Name

The Set DI Name is submenu screen under the main menu, the DI name can be self-defined by user according to the actual needs. The default alarm name can be viewed as **10.10. User Interface Board**.

11.14. Rectifier Setting

Table 11.14.1 Rectifier parameter setting description

Parameter name	Setting range	Default value	Setting description
System Rect Num	1-20	6	The default number of rectifier modules. It is recommended to set the maximum number of module slots in the power system. Notice: The sum of rectifier module and solar module cannot be more than 20.
System Solar Num	1-20	0	The default number of rectifier modules. It is recommended to set the maximum number of module slots in the power system. Notice: The sum of rectifier module and solar module cannot be more than 20.
Start Mode	Walk-In， Immed Mode， Pocl	Pocl	1. Walk-In: The rectifier or solar output voltage slowly rises to the floating charging voltage. 2. Immed Mode: The rectifier output voltage immediately rises to the floating charge voltage. 3. Pocl: The controller controls the output voltage to rise slowly from 42V to the floating charging voltage. This startup mode is recommended.
Walk-In Time	0~200s	12s	Rectifier module output voltage slowly rising time.
Start Volt	42~54V	42V	The rectifier module startup voltage
Rect. VoltUp	58.5-60.5V	59.5V	Set the overvoltage protect point of the rectifier module. Be careful, this value is not generally recommended to change.

11.15. Set Network Address

Table 11.5 Network parameter setting description

Parameter name	Default value	Setting description
IP Address	192.168.70.2	Set according to your need.
Netmask Address	255.255.255.0	
Gateway Address	192.168.70.1	
Trap Address1	0.0.0.0	The host address 1 of SNMP active report alarm. Set according to your need.
Trap Address2	0.0.0.0	The host address 2 of SNMP active report alarm. Set according to your need.

11.16. Set Diesel Generator

Table 11.16.1 DG parameter setting description

Parameter name	Setting range	Default value	Setting description
Start DG			
DG Volt Control	Enable, Disable	Enable	When set to ‘Enable’, the system will judge whether start DG according to system output voltage.
DG Start Volt	42V ~ 56V	48.5V	Start DG when the system output voltage is lower than the setting value
DG Cap Control	Enable, Disable	Disable	When set to ‘Enable’, the system will judge whether start DG according to battery remaining capacity;
DG Start Cap	20% ~ 80%	40%	When the ‘DG Cap Control’ is set to ‘Enable’, and the current battery remaining capacity is lower than the ‘DG Start Cap’, the DG will be started
DG Daily Control	Enable, Disable	Disable	When set to ‘Enable’, the system will judge whether start DG according to the daily starting time.
Daily Start DG On	00:00~23:59	17:00	When the ‘DG Daily Control’ is set to ‘Enable’, start DG at 17:00 PM every day by default.
DG Monthly Control	Enable, Disable	Disable	When set to ‘Enable’, the system will judge whether start DG according to the monthly starting time.
DG Monthly Day1	1~28	15	When the ‘DG Monthly Control’ is set to ‘Enable’, start DG at 15:00 PM on the 15th of every month by default. Daily monthly day can be set up to 2 different dates, means starting DG twice a month. Daily monthly day can also be the same, means starting DG once a month, users can choose freely.
DG Monthly Day2	1~28	15	
Monthly Start DG On	00:00~23:59	15:00	
Stop DG			
DG Stop Cap	85% ~ 100%	99%	When the ‘DG Cap Control’ is set to ‘Enable’, and the current battery remaining capacity is higher than the ‘DG Stop Cap’, the DG will be stopped.
DG Curr Control	Enable, Disable	Disable	When set to ‘Enable’, the system will judge whether stop DG according to battery current.
DG Stop Curr	1 ~ 50A	5A	When the ‘DG Curr Control’ is set to ‘Enable’, and the battery current is lower than the ‘DG Stop Curr, the DG will be stopped.
DG Stop Delay	0~1440min	5min	When the ‘DG Stop Cap’ or ‘DG Stop Curr’ condition is satisfied, stop DG after the delay time arrives

Monthly Charge Time	1~24H	2H	After monthly start DG, if the 'DG Stop Cap' or 'DG Stop Curr' condition is satisfied, DG will not be stopped until monthly charge time reaches the set value.
Low Fuel Stop DG	Enable, Disable	Disable	When set to 'Enable', the system will judge whether stop DG according to "Diesel Cap Alarm".
DG Max run time	1 to 10000 min	240min	DG will run for this set time
DI/DO Config for DG			
DG1-DI	None, DI1~DI8	DI7	Digital Input x is used to detect the running state of DG.
DG1-DO	None, DO1~DO8	DO7	Digital output x is used to start or stop DG.
DG2-DI	None, DI1~DI8	None	No digital Input is used to detect the running state of DG.
DG2-DO	None, DO1~DO8	None	No digital output is used to start or stop DG.
Level Num Setting			
Level TNum	0~18	0	Total number of diesel capacity levels
Total Cap Setting			
Total Cap	0~10000L	100L	Total diesel capacity
Diesel Cap Level			
Zero Level(L)	0~10000L	0L	Diesel capacity level zero(L)
Zero Level(V)	0~5V	0V	Diesel capacity level zero(V)
Full Level(L)	0~10000L	100L	Diesel capacity full level(L)
Full Level(V)	0~5V	5V	Diesel capacity full level(V)
x Level(L)	0~10000L	0L	Diesel capacity level 1~18(L)
x Level(V)	0~5V	0V	Diesel capacity level 1~18(V)
Diesel Cap Alarm			
Diesel Cap_H	50~100%	90%	High diesel capacity
Diesel Cap_L	0~50%	10%	Low diesel capacity

11.17. DG power limit

Refer the below table to set DG power limit on GP2600 controller for different DG capacity & with DG different loading conditions.

DG Rating in kVA	DG Power limit in Watt with 3KW /4KW Rectifier & 100% DG Loading – GP2600- SW Ver: 9.02150208
15	12000
20	16000
22	17600
30	24000
45	36000
60	48000

12. ATS Controller (optional)

Control card will continuously monitor the Mains /Grid power supply and DG power supply provides power relays for controlling Mains & DG Power

Control cards are built in High Voltage protection, Low voltage protection & surges protection.

ATS controller working logics

1. Grid healthy range is: 90 to 300VAC
2. DG Healthy range is: 90 to 300VAC
3. The controller continuously monitors both source Grid & DG Input healthy & priority will be for Grid in case grid is unhealthy second priority will be for DG source.
4. While the system working on DG mode if grid restored & healthy controller will change over to grid mode.
5. In case of emergency input high controller will disconnect both sources even input source is healthy

13. REMOTE MONITORING SYSTEM

GP Sonic is a 4-in-1 device – Remote Site Monitoring, GPS position, Wi-Fi connectivity for configuring & Ethernet local monitoring in one product. Remote configuration of GP Sonic is possible through Over the Air (OTA) process via server through SMS commands. Site parameters like data, alarms, Fuel, GCU parameters and GPS coordinates are sent network to central monitoring station via GPRS. GP Sonic automatically detects mode changes, alerts & periodic/request data accordingly push to the server.

13.1. Specification

Sl.No.	Parameters	Value
1.0	**Power Supply**	
1.0.1	Nominal & Operating voltage range	48V DC & 30-70Vdc
1.0.2	Power Consumption	Typical 12W
1.1	**System Specification**	
1.1.1	Controller Technology	Microprocessor Based
1.1.2	Modem	Quad Band 4G/3G/2G with auto fallback
1.1.3	SIM Module	Dual SIM (Preferred to go with micro footprint SIM)
1.1.4	GPS	GPS/GLONASS (Optional)
1.1.5	LED Indications	4 LED indications (System Run, Server Communication, Slave device Communication & GPRS Status)
1.1.6	Digital Input	16 DI's
1.1.7	Digital Output	8 DO's (Default NO)
1.1.8	Analog I/P	3 Analog inputs, one Voltage 0 ~60VDC, one Temperature measurement & one Current input 4~20mA (Optional)
1.1.9	Periodicity	120-86400 seconds (default 1800 Seconds)

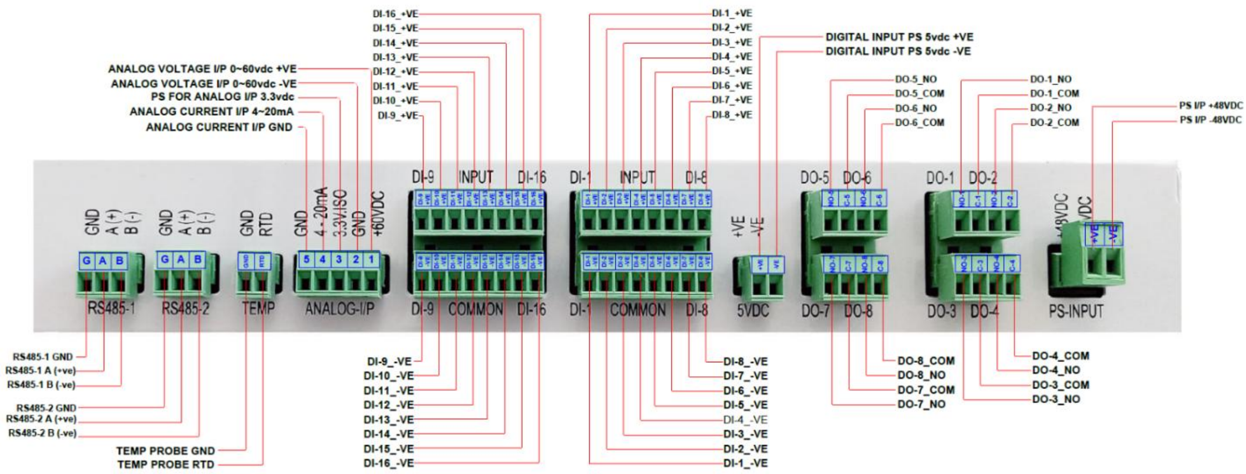
1.3	**Environmental Working Condition**	
1.3.1	Operational Temperature	-25°C to 55°C
1.3.2	Storage temperature	-30°C to 75°C
1.3.3	Humidity	95% RH (Non-Condensing)
1.3.4	Vibration Standards	BS6861: Part 1 Grade 2.5 for minimum vibration in operation
1.4	**Communication Ports & Protocol**	
1.4.1	RS485	2 RS485 ports (default connected to RS485-1 port)
1.4.2	Ethernet port	Debugging Console logs /Programming
1.4.3	GPRS	4G/3G/2G with auto fallback
1.4.4	Network Protocols:	HTTP & NTP
1.4.5	Configuration	OTA from server/SMS/Ethernet
1.4.6	Communication Types	Immediate – Alarm/Status change/ Request Scheduler – Configurable (120- 86400 seconds)
1.4.7	USB Host	Downloading history data

Note: Some of the above functions are optional and clients need to be informed in advance to factory about their requirements to configure as enable with appropriate accessories support

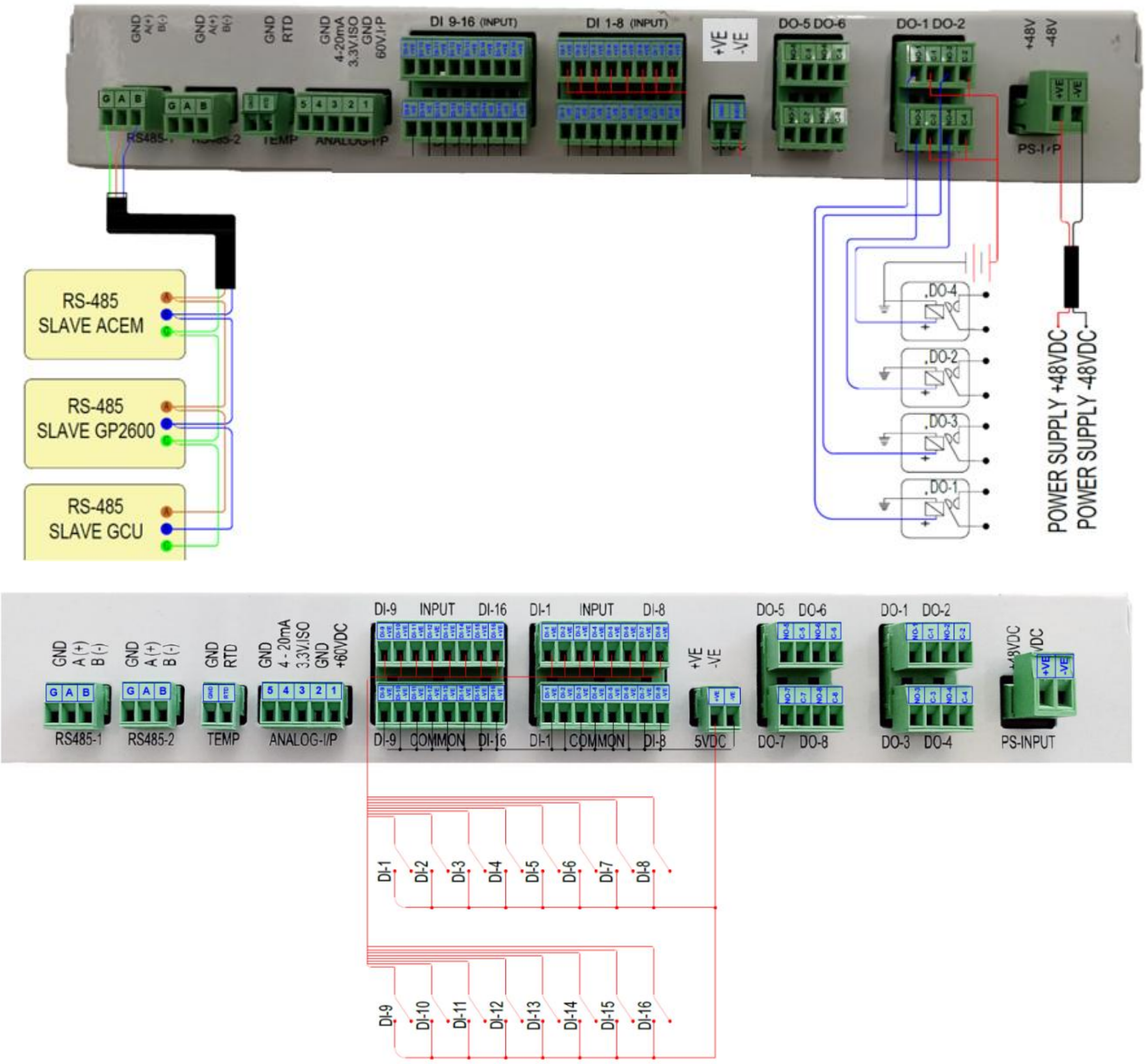
13.2. Installation Details

- Please ensure to install RMS in a place where sufficient data speed (minimum – 236kbit/s with end-to-end latency of less than 150ms) is available to communicate with the server.
- Make sure the SIM recharges and has the required data balance before installing it on the RMS.
- The antenna should be properly fixed outside of the system.

13.2.1. Sonic Installation:



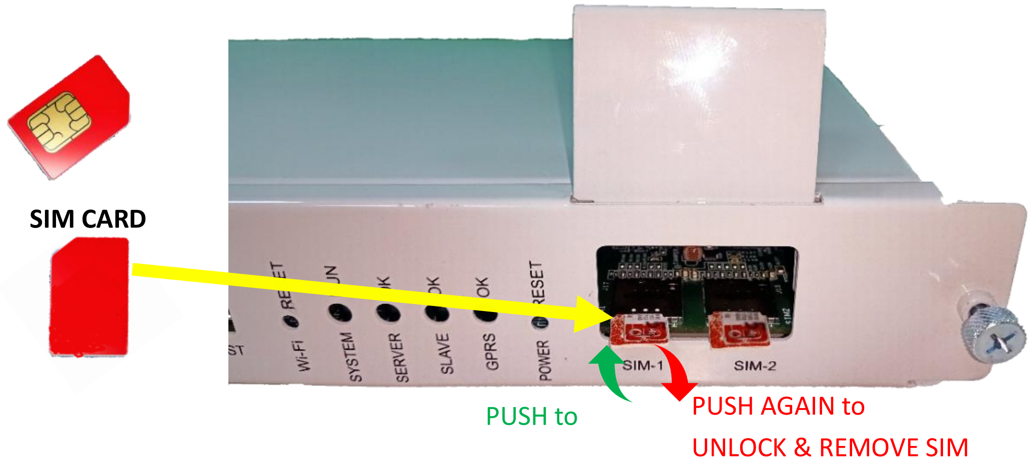
Generic wiring termination details of GP-SONIC RMS Picture:13.2.1



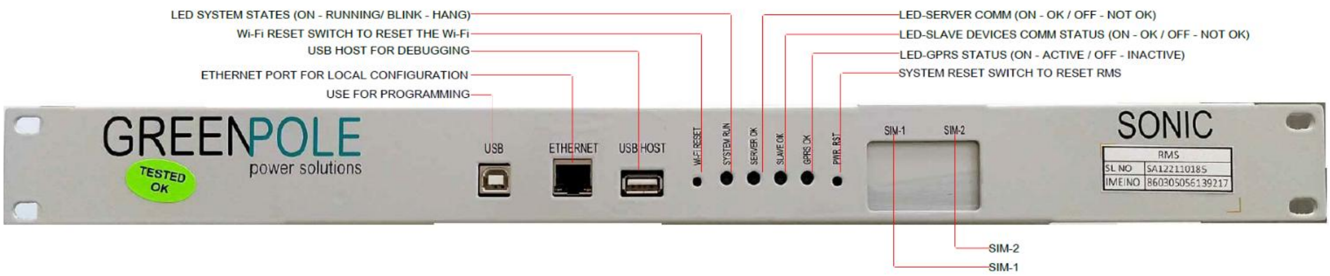
RMS DIGITAL OUTPUT	DEFAULT CONFIGURATION
DO-1	Smoke/Fire alarm
DO-2	Load On DG Alarm
DO-3	Fan Fail Alarm
DO-4	Door Open Alarm

Sonic wiring details Picture:13.2.2

Sonic Antenna installation details Picture:13.2.3



Sonic SIM Installation Picture:13.2.4



Sonic Font view with detail description Picture:13.2.5

NOTE: Please follow the installation steps as per below.

1. **Connect Digital input & Digital Output connectors.**
2. **Connect RS-485 communication connector**
3. **Connect GPRS, GPS & Wi-Fi Antenna & place the antennas properly near to cable glands or outside the cabinet to get proper network signal & route the antenna wires away from hot place & power cables**
4. **Connect Power supply connector & verify device is working properly**
5. **Once power on completed insert the SIM & configure Site ID and other parameters as per configuration guide Point-10**
6. **Check the device (GP-SONIC) communication status at server**
7. **Verify & configure right APN as per installed SIM operator**

13.3. SONIC Configuration & Commissioning

The following configurations can be configured by sending SMS commands

- Site ID Configuration
- Mobile Number Configuration
- APN configuration as per the installed SIM operator
- Upgrade Latest Firmware file

- Upgrade all Latest Library & kernel files
- Upgrade Configuration file
- Make sure all the slave devices (ACEM & GP2600) are with latest software version
- Set Push Frequency as per requirement (Min 30Min to avoid loading the server network)
- Reboot the Device
- Check Live status

NOTE	**All SMS commands need to be sent from whitelisted (pre-configured) number to site mobile (SIM) number.** **System will not accept any commands from non-whitelisted (pre-configured) numbers.** **In case whitelisted number is moving out/inactive need to be removed from linked systems and re-configured with new number.**

13.3.1. Configuring Site ID Using SMS Command

Configure the allotted site-ID in the Sonic RMS controller allowed up to 20 character it can be alpha numeric including special character except space & underscore.

SMS Command format: SET<space>SITE-ID<space>XYZ. To site SIM mobile number

Ex: **SET SITE-ID TEST-GREENPOLE_1**

13.3.2. Configuring Mobile Number-1 (SIM-1) Using SMS Command

Configure the Mobile number into Sonic RMS controller which is installed at the site in the predefined format (Country code followed by SIM calling number)

SMS Command format: SET<space>MOBILE-NO<space>+911234567890

Ex: **SET MOBILE-NO +911234567890**

13.3.3. Configuring Mobile Number-2 (SIM-2) Using SMS Command

Configure the Mobile number into Sonic RMS controller which is installed at the site in the predefined format (Country code followed by SIM calling number)

SMS Command format: SET<space>MOBILE-NO-2<space>+911234567890

Ex: **SET MOBILE-NO-2 +911234567890**

13.3.4. Configure APN Name as per installed SIM's Operators

Configure the installed SIM operator APN into Sonic RMS controller which is installed at the site in the predefined format by SMS Command format:

For SIM-1: SET<space>APN-1<space>APN name as per installed SIM-1 operator

Ex: **SET APN-1 airtelgprs.com**

For SIM-2: SET<space>APN-2<space>APN name as per installed SIM-2 operator

Ex: **SET APN-2 airtelgprs.com**

13.3.5. Upgrade System Firmware

System firmware/Software can be upgraded remotely (FOTA_ Firmware Over The Air) by sending SMS command to site mobile number if required.

SMS Command format: SET<space>FWIMAGE<space>SERVER-ADDRESS

Ex: **SET FWIMAGE SERVER ADDRESS**

13.3.6. Upgrade System Configuration file

System Configuration file can be upgraded remotely (OTA_Over The Air) by sending SMS command to site mobile number if required

SMS Command format: SET<space>CONFIG<space>SERVER-ADDRESS

Ex: **SET CONFIG SERVER ADDRESS**

13.3.7. Change Push Frequency

System to server communication (periodic data sending) frequency can be change by sending SMS command to site mobile number if required

SMS Command format: SET<space>PUSH-FREQUENCY TIME (in seconds)

Ex: **SET PUSH-FREQUENCY 1800**

13.3.8. Reboot the Device

To reboot/restart the device remotely send below SMS command to site mobile number

SMS Command: **SET SYSRESET**

13.3.9. Check Device Live Status

Send below SMS command to site mobile number to check the system live status (following information will be getting via SMS reply to Platform name, firmware version, device Uptime after reboot)

SMS Command: **GET STATUS**

13.3.10. Get Live Data

Send below SMS command to site mobile number to push live data to server (Live site data will be sent to server)

SMS Command: **GET LIVE-DATA**

13.4. LED's indication status

LED DESCRIPTION	LED STATUS	FUNCTION CONDITION
System Run		ON – System is normal and working
		OFF – System is Off (No power input)
		BLINKING – System is ON but not working
Server Ok		ON – Server connection successful
		OFF – Server connection Failure
Slave Ok		ON – Slave device in communication
		OFF – Slave device not in communication
GPRS Ok		ON – GPRS connection established successfully
		OFF – GPRS connection Failure

14. DG Battery Charger (Optional)

DG battery charger works DC/DC, this charger provided to avoid DG battery discharge during good grid and site battery backup (DG idle for long time)

Specification:

1. Input DC 40 to 60V
2. Output: 12V 120W
3. Protection: Short Circuit, Output Over Voltage, Batt Reverse
4. Input and Output is Isolated.
5. Indications: Input ON, Float, Boost, Batt. Rev.

15. Commissioning and Outdoor Power Cabinet Testing

15.1. Installation

Check that all input and output circuit breakers and fuses are in off state. All input and output connection cables, signal wires, working ground wires and protective ground wires are firmly connected. Switch off the battery circuit breaker to ensure no-load startup, and measure that there is no short circuit between input and DC output bus of each circuit.

Check box and earthing grounding resistance is not higher than 5 Ω between components.

Refer the SLD and perform connection as per below sequence.

Install Earth/ Ground cable

Step 1: Insert the Earth/Ground cable through the cable gland, connect to IGB (Internal Grounding Busbar) and tight it.

Step 2. Connect the ground/earth cable from common ground/earth bus bar to DC +ve common bus bar

Step 3. Connect the ground/earth cable from common ground/earth bus bar to system body

Install DC output cable

Step 1: place the DC output cable. Insert the DC output cable through the Cable Gland into the DCDB tenant on the left and tight the gland to hold and secure entry.

Step 2. Fastening the DC output -ve cable to the DC output of the corresponding specification to MCB according to the actual load capacity.

Step 3. Fastening DC output +ve cable to the screw of RTN+ bus specifications.

Install the battery cable

DANGER

1. No smoking or sparking near the battery.
2. Turn off the battery breaker before installing the battery.
3. Comply with battery manufacturer's regulations and warnings. Use tools with insulated handles, otherwise the battery may burn out and personal injury may occur.
4. Wear goggles, rubber gloves and protective clothing during battery operation.
5. Remove conductive items such as watches, bracelets, and rings. If battery fluid enters the eye, flush with cold water for more than 15 minutes and seek medical attention immediately.

6. If battery acid touches skin or clothing, wash immediately with soap and water.
7. Do not use metal to contact two or more battery terminals at the same time. Otherwise, a transient short circuit can produce a spark or explosion.
8. Do not short circuit or reverse connect positive or negative battery terminals during battery installation.
9. When the current is too high, loose connections can lead to excessive voltage drops or battery burning.
10. Make sure the battery comes from the same manufacturer and the same model and has close voltages. Old batteries cannot be used together.

Step 1: place the battery cable. Insert the battery cable through the cable gland on the right side and tight it.

Step 2. Fastening the battery -Ve cable to the battery empty.

Step 3. Fastening the battery +Ve cable to the RTN+ bus.

Install AC input cable (Grid and DG)

DANGER

1. Make sure the front AC input is left open to an OFF state and place a prominent "do not operate" sign.
2. Put all MCB OFF before installing the cable.

Step 1: put the 230/400Vac three-phase four-wire Grid AC input cable through the cable Gland on the right side and mount it on the terminal blocks at the rear side.

Step 2: Fastening the AC input line to the corresponding AC input open and terminal.

Step 3: put the 230/400Vac three-phase four-wire DG AC input cable through the cable Gland on the right side and mount it on the terminal blocks at the rear side.

Step 4: Fastening the AC input line to the corresponding AC input open and terminal.

15.2. Inspection Installation

1. Check hardware installation
   1.1 Check that all screws are properly fastened (especially those used for electrical connections) and that the flat washer and spring washer are properly installed.
   1.2 Check that the rectifier module is fully inserted into each slot and locked correctly.
2. Check electrical connection
   2.1 Check that all MCB are OFF.
   2.2 Check that the flat washer and spring washer are securely installed on all the terminals, and that all OT terminals are complete and correctly connected.
   2.3 Check whether the battery is properly installed and whether the battery cable is properly connected.
   2.4 Check that the input and output power cables and ground cables are properly connected.
3. Check the cable installation
   3.1 Check the cable installation
   3.2 Check that all cables are lined up neatly and tightly connected to their nearest cables, without twisting or overly bending.
   3.3 Check cable label is correct, firm, and consistent.

15.3. Functional Testing

- When there is Main power supply (Grid) available, turn on the Grid input breaker, make sure input voltage is within healthy range and switch on rectifier input one by one after getting AC input available through ATS, measure output DC voltage.
- When measuring the positive and negative bus voltage with a multimeter, it shall slowly rise to the default float charging voltage and switch off the Grid input.
- Turn off Grid Input and check DG function
- Turn on DG input MCB, through controller setting parameter create battery voltage or SOC low to get DG start signal automatically. Verify DG switched on and power extended till rectifier through ATS. Positive and negative bus voltage is measured with a multimeter, should slowly rise to a default value i.e., float charging voltage value, then stop the DG.
- Observe whether the system parameters are abnormal through the LCD screen and set the system working parameters according to the correct operation methods and steps.
- After confirming that the system is working normally and the measured parameters meet the requirements, turn off PV input and AC input (DG and Grid), connect to the load, run the system, turn on the battery fuse, and then system enters the normal working state.
- After starting up, the system shall be closely monitored for 20min, and the monitoring information provided by the system shall be paid attention to. The warning signal shall be processed in time. If the situation is serious or the test needs to shut down the system, the shutdown sequence is as follows:
  - Switch off the load, battery and AC input (Grid/DG).

16. Fault Analysis and Treatment

16.1. AC Power Failure

Fault reason:

- The AC input power cable is faulty.
- The AC input circuit breaker is OFF.
- The mains grid is faulty.

Solution:

- Check whether the AC input cable is loose. If yes, secure the AC input cable.
- Check whether the AC input circuit breaker is OFF. If yes, handle the back-end circuit failure and then switch on the circuit breaker.
- Check whether the AC input voltage is lower than 50 Vac. If yes, handle the mains grid fault.

16.2. AC Input Overvoltage

Fault reason:

- The AC input overvoltage alarm threshold is not set properly on the monitoring module.
- The power grid is faulty.

Solution:

- Check whether the AC input overvoltage alarm threshold is properly set. If no, adjust it to proper value.
- Check whether the AC input voltage exceeds the AC overvoltage alarm threshold (300 Vac by default). If yes, handle the AC input fault.

16.3. AC Input Undervoltage

Fault reason：

- The AC input undervoltage alarm threshold is not set properly on the monitoring module.
- The power grid is faulty.

Solution:

- Check whether the AC input undervoltage alarm threshold is properly set. If no, adjust it to a proper value.
- Check whether the AC input voltage is below the AC undervoltage alarm threshold (90Vac by default). If yes, handle the AC input fault.

16.4. DC Output Overvoltage

Fault reason：

- The DC overvoltage alarm threshold is not set properly on the monitoring module.
- The power system voltage is set too high in manual mode.
- Rectifiers are faulty.

Solution:

- Check whether the DC overvoltage alarm threshold (58Vdc by default) is properly set. If no, adjust it to a proper value.
- Check whether the system voltage is set too high in manual mode. If yes, confirm the reason and adjust the voltage to normal after the operation.
- Remove the rectifiers one by one and check whether the alarm is cleared. If the alarm still exists, reinstall the rectifier. If the alarm is cleared, replace the rectifier.

16.5. DC Output Undervoltage

Fault reason：

- AC power failure occurs.
- The DC under voltage alarm threshold is not set properly on the monitoring module.
- The system configuration is not proper.
- The power system voltage is set too low in manual mode.
- Rectifiers are faulty.

Solution:

- Check whether an AC power failure occurs. If yes, resume the AC power supply.
- Check whether the DC under voltage alarm threshold (48.4Vdc by default) is properly set. If no, adjust it to a proper value.
- Check whether the load current is greater than the current power system capacity.
- If yes, expand the power system capacity or reduce the load power.
- Check whether the system voltage is set too low in manual mode. If yes, confirm the
- reason and adjust the voltage to a proper value after the operation.
- Check whether the power system capacity is insufficient for the loads due to rectifier
- failures. If yes, replace the faulty rectifier.

16.6. Battery Overcharged

Fault reason：

- The rectifier communication is interrupted
- Poor contact with the monitoring module.
- The monitoring module is faulty.

Solution:

- Check whether an alarm is generated for rectifier communication interruption. If yes, remove the rectifier and reinstall it to check whether the alarm is cleared. If the alarm still exists, replace the rectifier.
- Remove the monitoring module and reinstall it to check whether the alarm is cleared. If the alarm still exists, replace the monitoring module.

16.7. LLVD Disconnected

Fault reason：

- AC power failure occurs.
- Loads are manually disconnected.
- The load disconnection voltage is set too high on the monitoring module.
- Rectifiers are faulty.
- The system configuration is not proper.

Solution:

- Check whether an AC power failure occurs. If yes, resume the AC power supply.
- Check whether loads are manually disconnected. If yes, confirm the reason for the manual disconnection, and reconnect the loads after the operation.
- Check whether the load disconnect voltage (48Vdc by default) is set too high on the monitoring module. If yes, adjust it to a proper value.
- Check whether the power system capacity is insufficient for the loads due to rectifier failures. If yes, replace the faulty rectifier.
- Check whether the load current is greater than the current power system capacity. If yes, expand the power system capacity or reduce the load power.

16.8. BLVD Disconnected

Fault reason：

- AC power failure occurs.
- Batteries are manually disconnected.
- The battery disconnect voltage is set too high on the monitoring module.
- Rectifiers are faulty.
- The system configuration is not proper.

Solution:

- Check whether an AC power failure occurs. If yes, resume the AC power supply.
- Check whether batteries are manually disconnected. If yes, confirm the reason for the manual disconnection, and reconnect the batteries after the operation.

- Check whether the battery disconnection voltage (46.2Vdc by default) is set too high on the monitoring module. If yes, adjust it to a proper value.
- Check whether the power system capacity is insufficient for the loads due to rectifier failures. If yes, replace the faulty rectifier.
- Check whether the load current is greater than the current power system capacity. If yes, expand the power system capacity or reduce the load power.

16.9. High Amb. Temp.

This alarm is generated only for the power system that has ambient temperature sensors installed.

Fault reason：

- The ambient over temperature alarm threshold is not set properly on the monitoring module.
- The temperature control system is faulty in the cabinet where the ambient temperature sensor is located.
- The ambient temperature sensor is faulty.

Solution:

- Check whether the ambient temperature alarm threshold (55°C by default) is properly set on the monitoring module. If no, adjust it based on site requirements.
- Check whether the temperature control system in the cabinet is faulty. If yes, rectify the fault. The alarm is cleared when the cabinet temperature falls within the allowed range.
- Check whether the ambient temperature sensor is faulty. If yes, replace the ambient temperature sensor.

16.10. Low Amb. Temp.

This alarm is generated only for the power system that has ambient temperature sensors installed.

Fault reason：

- The ambient low temperature alarm threshold is not set properly on the monitoring module.
- The temperature control system is faulty in the cabinet where the ambient temperature sensor is located.
- The ambient temperature sensor is faulty.

Solution:

- Check whether the ambient low temperature alarm threshold (0°C by default) is
- properly set on the monitoring module. If not, adjust it based on site requirements.
- Check whether the temperature control system in the cabinet is faulty. If yes, rectify the fault. The alarm is cleared when the cabinet temperature falls within the allowed range.
- Check whether the ambient temperature sensor is faulty. If yes, replace the ambient temperature sensor.

16.11. Batt. High Temp.

This alarm is generated only for the power system that has battery temperature sensor installed.

Fault reason：

- The battery over temperature alarm threshold is not set properly on the monitoring module.
- The battery temperature controlling system is faulty.
- The battery temperature sensor is faulty.

Solution:

- Check whether the battery over temperature alarm threshold (55°C by default) is properly set. If not, adjust it to a proper value.
- Check whether the battery temperature controlling system is faulty. If yes, rectify the fault. The alarm is cleared when the battery temperature falls within the allowed range.
- Check whether the battery temperature sensor is faulty. If yes, replace the temperature sensor.

16.12. Batt. Low Temp.

This alarm is generated only for the power system that has battery temperature sensor installed.

Fault reason：

- The battery under temperature alarm threshold is not set properly on the monitoring module.
- The battery temperature controlling system is faulty.
- The battery temperature sensor is faulty.

Solution:

- Check whether the battery under temperature alarm threshold (0°C by default) is properly set. If no, adjust it to a proper value.
- Check whether the battery temperature controlling system is faulty. If yes, rectify the fault. The alarm is cleared when the battery temperature falls within the allowed range.
- Check whether the battery temperature sensor is faulty. If yes, replace the temperature sensor.

16.13. Door Alarm

This alarm is generated only for the power system that has door status sensor installed.

Fault reason：

- The cabinet doors are open.
- The door status sensor is faulty.

Solution:

- Close cabinet doors.
- Check whether the door status sensor is faulty. If yes, replace the door status sensor.

16.14. Smoke/Fire Alarm

This alarm is generated only for the power system that has smoke sensors installed.

Fault reason：

- There is smoke inside the cabinet.
- The smoke sensor is faulty.

Solution:

- Check whether there is smoke inside the cabinet. If yes, disconnect the power supply
- from the cabinet, handle the fault, and then resume system operation and clear the alarm on the monitoring module.
- Check whether the smoke sensor is faulty. If yes, replace the smoke sensor.

16.15. Rect. Fault

Fault reason：

- The rectifier is in poor contact.
- The rectifier is faulty.

Solution:

- Check the Fault indicator on the rectifier panel. If it is steady red, remove the rectifier, and then reinstall it after the indicator turns off.
- If the alarm still exists, replace the rectifier.

16.16. Rect. Protection

Fault reason：

- The rectifier input voltage is too high.
- The rectifier input voltage is too low.
- The ambient temperature is too high.
- The rectifier is abnormal.

Solution:

- Check whether the AC input voltage exceeds the upper threshold of the rectifier working voltage. If yes, rectify the power supply fault and then resume the power supply.
- Check whether the AC input voltage is below the lower threshold of the rectifier working voltage. If yes, rectify the power supply fault and then resume the power supply.
- Check whether the ambient temperature is higher than the normal operating temperature of the rectifier. If yes, check and rectify the temperature unit fault.
- Remove the rectifier that generates the alarm and reinstall it after the indicator turns off. If the alarm still exists, replace the rectifier.

16.17. Rect. Comm Fault

Fault reason：

- The rectifier is removed.
- The rectifier is in poor contact.
- The rectifier is faulty

Solution:

- Check whether the rectifier is removed. If yes, reinstall it.
- If the rectifier is in position, remove the rectifier and reinstall it
- If the alarm still exists, replace the rectifier.

16.18. AC SPD Alarm

Fault reason：

- The AC SPD is faulty.
- The AC SPD detection cable is disconnected.

Solution:

- Check whether the AC SPD indication window turns red. If yes, replace the SPD.
- Check whether the AC SPD detection cable is disconnected. If yes, reconnect the cable.

17. Identifying Component Faults

17.1. Identifying AC SPD Faults

- Check the color of the AC SPD indication window. Green indicates that the AC SPD is normal. Red indicates the AC SPD is faulty.

17.2. Identifying Circuit Breaker Faults

The following lists main circuit breaker faults：

- The circuit breaker cannot be switched to ON/OFF after the short circuit fault for its end circuit is rectified.
- When the circuit breaker is switched to ON and its input voltage is normal, the voltage between the two ends of the circuit breaker exceeds 1 V.
- The input voltage is normal, but the resistance between both ends of the circuit breaker is less than1kΩ when the circuit breaker is OFF.

17.3. Identifying Rectifier Faults.

A rectifier is damaged if any of the following conditions is not met：

- When the rectifier does not communicate with the monitoring module and the AC input voltage is around 220Vac, the green indicator on the rectifier is steady on, the yellow indicator is blinking, the red indicator is off, and the rectifier output is normal.
- The monitoring module can perform equalized charging, float charging, and current limiting control for the rectifier when the communication cable to the rectifier is correct.

17.4. Identifying Monitoring Module Faults

The following are the main symptoms of monitoring module faults：

- The DC output is normal while the green indicator on the monitoring module is off.
- The monitoring module breaks down or cannot be started. Its LCD has abnormal display or buttons cannot be operated.
- With the alarm reporting enabled, the monitoring module does not report alarms when the power system is faulty.
- The monitoring module reports an alarm while the power system does not experience the fault.
- The monitoring module fails to communicate with the connected lower-level devices while the communications cables are correctly connected.
- Communication between the monitoring module and all rectifiers fails while both the rectifiers and the communications cables are normal.
- The monitoring module cannot monitor AC or DC power distribution when communications cables are intact, and AC and DC power distribution is normal.
- Parameters cannot be set or running information cannot be viewed on the monitoring module.

18. System Maintenance

18.1. Routine Maintenance

Routine maintenance is to find out the potential causes of failure in time before the failure of the equipment affects the business and to carry out effective maintenance. Processing to avoid business impact. Maintenance engineers must be familiar with relevant site conditions before maintenance and be prepared with maintenance tools in advance. The system can work continuously and uninterruptedly. In order to ensure the reliability of power supply, it is necessary to do a good job in daily maintenance to make the power supply system work in the best state.

I. **Regular cleaning of the power supply system (Modules, air filters, door gasket and cable entry & exit) and photovoltaic module to prevent dust accumulation and other phenomena every week or 15 days once. (based on the site environmental condition)**

II. Check the parameters of the power system once a week if they are normal. If there are abnormalities, they should be eliminated in time. If there are alarms, the system should be dealt with on time.

III. Every month or after the thunderstorm should check whether the lightning protector is invalid, if it is invalid, it should be replaced in time.

IV. Check the fuse status once a month, and have a certain number of standby fuses, can be replaced in time when the failure occurs. Check the connection of input and output wiring loosening and bad contact once a month.

V. Check every month whether the input and output conductors are intact and intact. If there is any damage, they should be replaced in time. Check every circuit breaker once a half year whether it can switch flexibly, whether it burns or not.

VI. Maintenance and maintenance of batteries shall be carried out in accordance with the relevant instructions of batteries.

Table 17.1.1 List of Periodic Maintenance

Maintenance items	Maintenance content		
	Checkpoints	**Inspection methods**	**processing method**
Appearance of cabinet	No damage to locks of cabinet doors	Visual inspection, switch door lock	Lock changing
	Cabinet damaged and deformed	Visual measurement	If the damage is serious, the cabinet should be replaced.
	Is there corrosion and rust in cabinet	Visual measurement	Re-painting after rust removal
	Does the cabinet have dust, oil and other covering	Visual measurement	Clean with soft cotton cloth
Dustproof net	Are there any ash deposits	Visual measurement	Clean up
Temperature in cabinet	Is it too high or too low	Thermometer measurement	Take reasonable heating or cooling measures.

Fan	Is there any ash on the fan surface	dust	Dust removal
	Fan appearance, speed,	Is noise and vibration normal?	Replacement of fans
system output	Is Voltage Output Normal	A multimeter	Refer to Fault Analysis and Handling
indicator	if the indicator is normal	Visual measurement	Refer to Fault Analysis and Handling
Wireway	Whether the insulation layer and terminal are damaged or not	Visual measurement	Replacement of cables or insulated terminals

18.2. Circuit Breaker Replacement

Preparation in advance

- Prepare tools and materials: anti-static wristband/anti-static gloves, cross screwdriver, one-word screwdriver, anti-static box/anti-static bag, cabinet door key.
- Prepare the lightning arrester module or lightning arrester that needs to be replaced.

⚠ **Warning**

The front power source must be disconnected before the circuit breaker is replaced.

Operational steps

I. Wear anti-static wristband or gloves. And disconnect the power source of the front stage.

II. Remove the cables on the circuit breaker in turn with cross screwdrivers. Insulate the cables in time and label them in the process of disassembly.

III. Use a screwdriver to force down to empty the bottom contacts and then remove the empty contacts. The replacement method is shown in Figure 17.2.1.

IV. Put the new empty-opening card into the original position and tighten the wire.

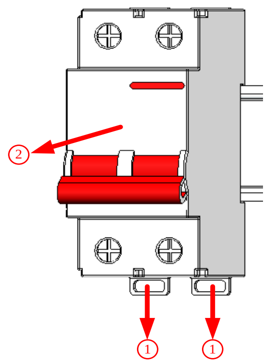


Figure 17.2.1 Circuit Breaker Replacement

18.3. Controller Replacement

Preparation in advance

- Prepare tools and materials: anti-static wristband/anti-static gloves, anti-static box/anti-static bag, cabinet door key. Prepare controllers to be replaced.

Operational steps

I. Wear anti-static wristband or gloves.

II. The controller has the function of hot swapping, unscrewing the loose screw at both ends of the controller and pulling the controller out. Insert the new controller and fasten the unscrewed screw. As shown in Figure 17.3.1.

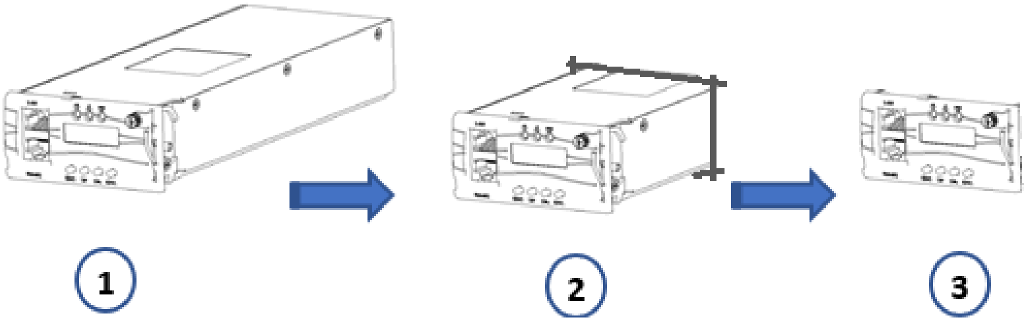


Figure 17.3.1 Controller Replacement

18.4. LVD Force on Switch Operation Instruction

1. In case of emergency (if LVD board fail while replacing LVD board) follow the below steps to operate.
2. First switch on the BLVD (SW-2) force on switch
3. Then switch on the LLVD (SW-1) force on switches.
4. After replacing LVD board Switch off the force on switches LLVD (SW-1) first then BLVD (SW-2) sequentially

19. Client Management

19.1. WEB GUI Instructions

Through the web page, users can achieve the following functions:

- View real-time data or status
- Send control commands
- Set parameters
- Pop-up the current alarm information automatically.
- Check history warning information and battery diary.
- Upload or download configuration files, historical alerts and historical events.

19.2. Setting up the Computer's Network Local Connection

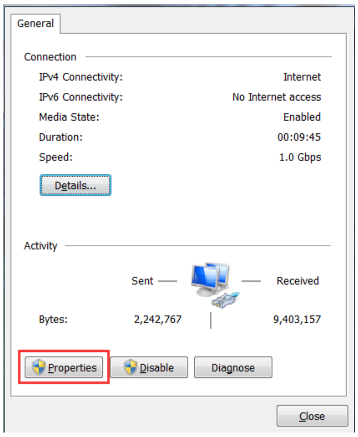
The monitoring module is equipped with a web server with a default IP address 192.168.70.2.

Notice: It is possible to change the web server's IP address in display menu Parameter settings/Comm Settings / IP/Subnet/Gate.

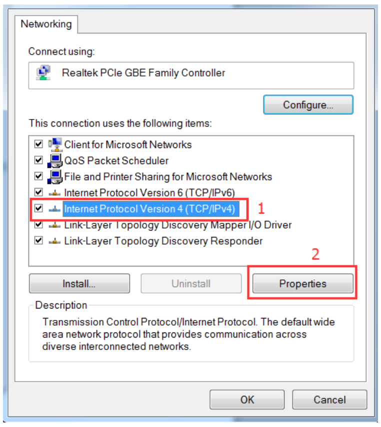
The monitoring module web server can be connected to a PC:

- Directly by using a crossed type of network cable
- Through a LAN

1) Open the computer's network local connection and click the "Properties" button:

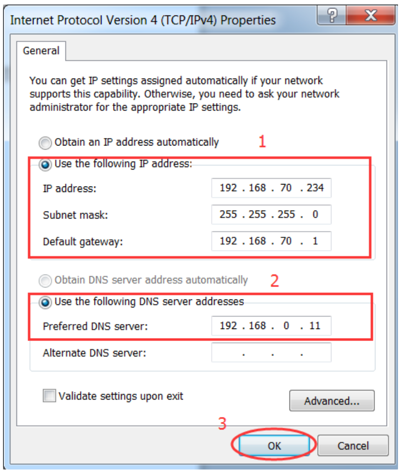


2) Double-click the "TCP/IPv4" option:



3) The IP address is set to 192.168.70.X, where X represents any number from 3 to 254; the subnet mask, default gateway settings are as follows; After setting, click "OK".

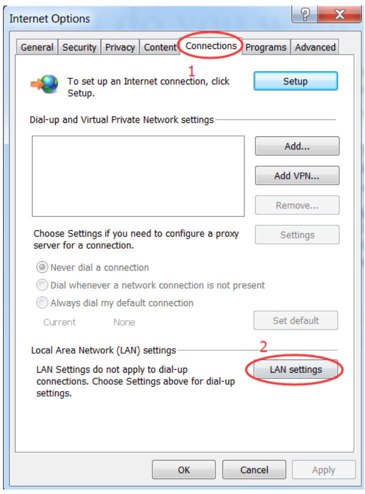
Notice: the DNS server can ignore settings; it is not needed.



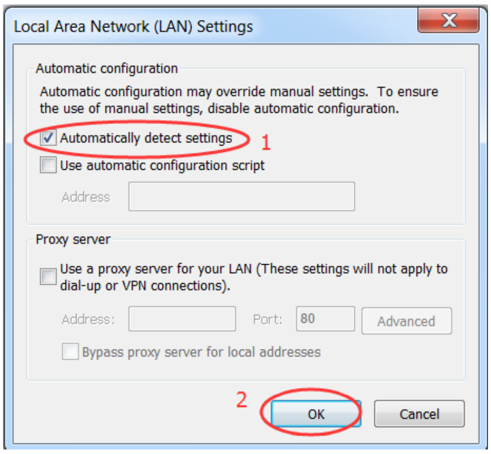
19.3. Setting Up the "Internet Explorer" Web Browser

1) Launch "Internet Explorer".

2) Select Internet Options from the Tools menu. The "Internet Options "window opens. In the "Internet Options "window, select the Connections tab.GP2600



3）Click on the LAN Settings... button. The following window opens. In the LAN Settings window, uncheck the proxy box and click OK.



19.4. Login Page.

In “Internet Explorer”, enter the default IP address (192.168.70.2) programmed into the controller and press ENTER. The following WEB Interface window opens.

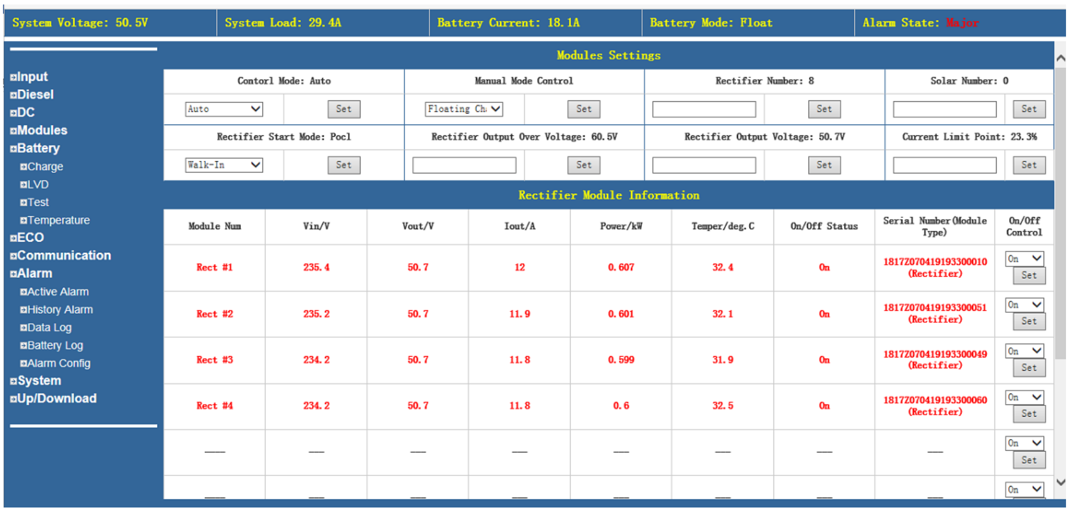
Enter a valid Username and Password, then click OK. By default, there are two “Username” and “Password” combinations. Credentials will be shared separately to authorized person.

The username of “admin” has the highest authority, and the username “operator” has no authority for uploading and downloading configuration files.

19.5. WEB GUI

This WEB GUI provides a full function to customer. Via a PC connected internet, you can control the power system, view the working status, change the parameters, and download the Alarm history record.

WEB GUI screens display as follows:



20. Warranty Exclusion & Recommendation

20.1. Conditions or scenarios where the warranty does not apply, such as:

- Improper handling, transportation, installation, use or maintenance.
- Damage due to accidents, natural disasters or environmental conditions (Flod, Earth quack, Fire, High wind etc.) outside the specific/product range.
- Unauthorized repairs or modifications, addition of external equipment and tempering cases.
- Abnormal input to the system and related product.
- Costs that may be incurred by the customer, such as shipping or handling fees or special request to support.

20.2. Climatic and Environmental Conditions

Dust and Particulate Matter:

- Adopt proper and routing maintenance for dust and particulate levels in the operating environment.
- Necessity of filters, enclosures, Rack & Rectifier cleaning to protect the Power system and related product

Recommendations:

- Regular Maintenance-Weekly basis to ensure optimal performance and longevity.

Corrosive Environments:

- Restrictions on operating the Power System in corrosive environments.
- Recommend protective measures or coatings if necessary.

Recommendations:

- Regular Maintenance-Weekly basis to ensure optimal performance and longevity.

Electrical Environment:

- Requirements for input voltage, frequency, and power quality must be as specified on product and manual.
- Surge protection and grounding maintained as recommended.

Operating Temperature Range:

- Minimum and maximum operating temperatures must be as specified (e.g., 0°C to 50°C).
- Temperature-related conditions that could affect warranty coverage.

Altitude:

- Maximum operating altitude must be withing range (e.g., up to 2000 meters above sea level).

20.3. Additional Recommendation

Record Keeping:

Keeping detailed records of installation, maintenance, and any issues encountered to support potential warranty claims.

21. ODR Power Cabinet I&C Acceptance Report

DC POWER SYSTEM CHECK LIST	
CUSTOMER:	**SITE IN CHARGE NAME & NUM:**
SITE NAME/CODE:	**INSTALLATION DATE:**
COMMISSIONING DATE:	**POWER SYSTEM SR NO:**
CONTROLLER SR. NO:	**CONTROLLER VERSION:**
I&C ENG. NAME:	**CONTACT NO:**
RECTIFIER MODULE SERIAL NO:	
1.	7.
2.	8.
3.	9.
4.	10.
5.	11.
6.	12

VISUAL INSPECTION/OBSERVATIONS				
SL.NO	**PARAMETER**	**OBSERVATION STATUS**		**REMARKS**
1	MOUNTING OF RACK/SUB RACK MECHANICAL NUTS / BOLTS	OK	NOT OK	
2	TIGHTNESS OF ELECTRICAL TERMINATION	OK	NOT OK	
3	LUGGING & CRIMPING OF INSTALLATION CABLES	OK	NOT OK	
4	FIXING OF NUTS & BOLTS AT MOUNTING HOLES	OK	NOT OK	
5	POWER & CONTROL CABLE ROUTING	OK	NOT OK	
6	TERMINATION OF BATTERY CABLE +VE BUS BAR & BATT FUSE	OK	NOT OK	
7	TERMINATION OF LOAD MCB'S & BUS BAR +VE CABLES	OK	NOT OK	
8	RECTIFIER PLACING IN THE RACK	OK	NOT OK	
9	TERMINATION EARTHING CABLE FOR MAINS & DC BUS BAR	OK	NOT OK	
10	FIXING OF CABLE GLANDS/GRUMMET	OK	NOT OK	
Sl. No	**PARAMETERS**	**DISPLAY VALUE**		**MEASURED VALUE**
1	GRID/AC INPUT VOLTAGE		Vac	
2	GRID/AC INPUT CURRENT		Aac	
3	DG INPUT VOLTAGE		Vac	
4	DG INPUT CURRENT		Aac	
5	BATTERY VOLTAGE		Vdc	
6	DC OUTPUT VOLTAGE & CURRENT	Vdc.	Adc	
7	SET BATTERY CAPACITY		Ah	Installed
8	BATTERY CHARGE CURRENT LIMIT		Amp. Dc	
Sl. No	**ALARM**	**PFC TEST (OK/NOT OK**		**DISPLAY VALUES**
1	LOAD MCB TRIP / LLVD OPEN	OK	NOT OK	
2	AMBIENT/BATTERY TEMPERATURE LOW/ HIGH	OK	NOT OK	
3	BATTERY DISCONNECT/FUSE FAIL/BMS COM FAIL	OK	NOT OK	
4	AC INPUT VOLTAGE LOW/HIGH	OK	NOT OK	
5	SYSTEM DC VOLTAGE LOW/HIGH	OK	NOT OK	
6	SINGLE/MULTIPLE RECTIFIER FAIL	OK	NOT OK	

7	DG REMOTE START SIGNAL	OK	NOT OK	
8				
BATTERY BANK				
BATTERY MODEL/TYPE:			BATTERY AH:	
NUMBER OF BATTERIES INSTALLED:			BATTERY TEMP SENSOR AVAILABLE (YES/NO):	
SL.NO	PARAMETER		STATUS, PLEASE TICK MARK (√)	
1	VISUAL CHECK OF BATTERY AGAINST DAMAGES, LEAKAGE & ABNORMALITIES		OK	NOT OK
2	MOUNTING/ASSEMBLY OF INDIVIDUAL BATTERY MODULE		OK	NOT OK
3	CONNECTION OF +VE & -VE TERMINAL OF BATTERY FROM POWER CABINET UNIT		OK	NOT OK
4	MEASUREMENT OF INDIVIDUAL BATTERY MODULE VOLTAGE (PLEASE HIGHLIGHT IF ANY DAMAGES OR VOLTAGE UNHEALTHY		OK	NOT OK
5	VOLTAGE MEASUREMENT OF TERMINAL w.r.t EARTH		OK	NOT OK
6	BATTERY SERIAL NUMBER		OK	NOT OK
7	OPEN CIRCUIT VOLTAGE		OK	NOT OK
GREENPOLE POWER SOLUTIONS		**CUSTOMER**		
NAME:		**NAME:**		
DESIGNATION:		**DESIGNATION:**		
CONTACT NO:		**CONTACT NO:**		
SIGNATURE & DATE:		**SIGNATURE & DATE:**		